

PCR-7

Observable Projection and Experimental Prediction Lock for the Neutrino Carrier

Charged-current closure invariants, finite-protocol reference separation, and held-out falsification after neutrino carrier identity



Scope of the result

The theorem applies to the two source-qualified vectorlike-joint sectors completed by PCR-5E and characterized by PCR-6. It identifies a common charged-current closure packet, establishes a finite flavor-complete spectral-tomography protocol with uniform positive acceptance and stratified nonlinear identifiability, and proves positive confidence-region separation from independently formed no-vectorlike reference theories. The protocol and falsifier are fixed before held-out validation records are examined. The result is a target-qualified experimental prediction, not a claim of current empirical confirmation, present-facility completion, or universal validity beyond the two completed sectors.

Abstract

PCR-6 establishes that every realization in two completed vectorlike-joint neutrino sectors contains one terminal continuation-bearing carrier with three nondegenerate light standing sectors and one rank-three degenerate heavy standing sector. The present paper determines whether that target-qualified architecture entails an experimental record that remains unavoidable after all lawful realization, parameter, matching, detector, and same-child decoupling freedom has been included.

The empirical roles are fixed before the observable analysis: records already used as target anchors, calibration inputs, or derivation data are excluded from subsequent validation. Observable completeness is formulated semantically through target-admitted source, protocol, detector, nuisance, and record uses, and the no-vectorlike reference is formed independently while preserving the same charged-current boundary, calibration system, background class, and detector quotient.

For every accepted realization, the light and heavy charged-current blocks have the form

$$N = c_E U, \quad V_H = -s_E U, \quad \delta_E = s_E^2 > 0,$$

with U unitary and a single threefold-degenerate heavy pole $H_\lambda \in H(I_E)$. Define

$$\Delta_{\text{act}} := \mathbf{1}_3 - NN^\dagger, \quad \Omega_H := V_H V_H^\dagger, \quad C_H := \Delta_{\text{act}} - \Omega_H,$$

and augment the packet by the heavy-pole support class and heavy GIM/CLFV residue. The source identities imply

$$\Delta_{\text{act}} = \Omega_H = \delta_E \mathbf{1}_3, \quad C_H = 0, \quad \text{rank } \Omega_H = 3, \quad \text{Pole}_H = \text{Deg}_3(H(I_E)), \quad \mathcal{A}_{\mu e}^H = 0.$$

These relations are invariant under every lawful source, gauge, basis, matching, continuation, and groupoid morphism. They are independent of the absolute light scale, the D1/D4 enriched scale, the accidental D2 amplitude, and the accidental/exact-global target decoration, and therefore define one common response class on both completed sectors.

A finite flavor-complete charged-current spectral-tomography protocol connects this source-level packet to detector records. Tagged charged-current parent events provide light normalization and appearance records, inclusive heavy production in every boundary flavor, all kinematically applicable correlated heavy production-decay or scattering records, and a finite scan of the compact heavy interval. Carrier and reference responses share the same calibration, sideband, mass-response, and reconstruction system. Exact heavy degeneracy makes the unresolved heavy response depend on $V_H V_H^\dagger$, while the measurable null geometry is expressed in the invariant squared-modulus coordinates $|(V_H V_H^\dagger)_{\alpha\beta}|^2$ rather than in an ill-defined linear amplitude at zero.

The detector quotient is consequently stratified. Diagonal Gram coordinates receive two-sided confidence intervals, whereas predicted off-diagonal nulls receive one-sided upper bounds on squared-magnitude coordinates. Uniform positivity of the physical response is established before compactness is used to select a finite cover. Finite predeclared exposures then yield

$$|\widehat{d}_\alpha - d_\alpha| < \frac{\delta_E}{8}, \quad q_{\alpha\beta}^U < \left(\frac{\delta_E}{16}\right)^2,$$

with $O(\delta_E^{-2})$ scaling for diagonal coordinates and $O(\delta_E^{-4})$ scaling for null-amplitude certification. Gershgorin's theorem gives rank three, and the induced positive-semidefinite packet confidence class has radius

$$\varepsilon_\star = \frac{3\delta_E}{8} < \frac{\sqrt{6}}{2}\delta_E.$$

The theorem therefore establishes a finite in-principle measurement protocol without asserting that a present facility has already accumulated the required exposure.

The independently formed references satisfy

$$\Delta_{\text{act}}^0 = \Omega_H^0 = 0, \quad \text{rank } \Omega_H^0 = 0, \quad \text{Pole}_H^0 = \text{NoHeavyPole}.$$

The carrier and reference classes are separated by their matrix and pole structure. Their ideal continuous-coordinate distance is $\sqrt{6}\delta_E$, and the reconstructed confidence classes retain the positive gap

$$\sqrt{6}\delta_E - \frac{3\delta_E}{4} > 0.$$

No lawful same-child path reaches the reference: s_E is fixed by the source-qualified certificate, the heavy pole remains inside the completely scanned compact interval, unitary changes cancel in the Gram matrices, and loss of acceptance or identifiability leaves the certified protocol rather than defining a nuisance limit.

The resulting status is

CrossChildJointPacketPredictionLock_{CCST}

for the two completed vectorlike-joint sectors, with empirical status

FrozenPredictionAwaitingHeldOutData

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The minimal locked packet consists of the active-deficit matrix, inclusive heavy-weight matrix, and heavy-pole support class. Existing oscillation and D-series records retain their roles as inputs; the flavor-complete heavy-inclusive record remains reserved for held-out validation. No conclusion is transported to unresolved PCR-5E parent sectors.

Keywords. AASC; neutrino carrier; observable projection; prediction lock; heavy neutral state; nonunitarity; charged-current tomography; quadratic null certification; reference separation; decoupling; held-out validation.

Principal theorem

Cross-child charged-current closure prediction

Let

$$\mathcal{B}_7^{\text{ready}} = \{b_{\text{acc}}, b_{\text{glob}}\}$$

be the two PCR-5E/PCR-6 ready child scopes. There exists a finite, source-authorized, pre-registered tagged charged-current protocol $p_\star$ and a detector-record packet

$$O_\star = (\Delta_{\text{act}}, \Omega_H, C_H, \text{Pole}_H, \mathcal{A}_{\mu e}^H)$$

such that, for every $b \in \mathcal{B}_7^{\text{ready}}$, every $X \in \text{PhysMod}_{E, \text{VJ}}^b$, and every lawful child continuation,

$$O_\star(X) \in \mathcal{R}_\star := \{(\delta_E \mathbf{1}_3, \delta_E \mathbf{1}_3, 0, \text{Deg}_3(H), 0) : H \in H(I_E)\}.$$

For every independently frozen no-vectorlike-carrier reference R_0 in the reference family,

$$O_\star(R_0) \in \mathcal{R}_0 := \{(0, 0, 0, \text{NoHeavyPole}, 0)\}.$$

The source-to-record protocol has a derived uniform acceptance $a_\star > 0$, an injective finite response map on the stratified diagonal/quadratic/pole quotient, common carrier/reference calibration and background controls, and finite exposures producing a simultaneous confidence relation of coverage at least $1 - \alpha_\star$ with

$$|\hat{d}_\alpha - d_\alpha| < \frac{\delta_E}{8}, \quad q_{\alpha\beta}^U < \left(\frac{\delta_E}{16}\right)^2.$$

These bounds imply rank three and place both reconstructed continuous packet classes inside balls of radius

$$\varepsilon_\star = \frac{3\delta_E}{8}.$$

The two ideal classes are disjoint under the exact rank/pole separator and satisfy

$$\text{dist}(\mathcal{R}_\star, \mathcal{R}_0) = \sqrt{6} \delta_E.$$

On the joint coverage event,

$$\text{dist}(\mathcal{D}_{p_\star}(\mathcal{R}_\star), \mathcal{D}_{p_\star}(\mathcal{R}_0)) \geq \sqrt{6} \delta_E - \frac{3\delta_E}{4} > 0.$$

The separation is nondecoupling throughout the lawful child closure. Therefore

$$\text{Status}(\text{PCR-7}; \mathcal{B}_7^{\text{ready}}) = \text{CrossChildJointPacketPredictionLock}_{\text{CCST}}.$$

The validation status is `FrozenPredictionAwaitingHeldOutData`, not empirical confirmation or current-facility completion.

Scope and interpretation

Domain of validity

The theorem is relative to the accepted PCR-5E and PCR-6 child certificates, including the fixed positive source coefficient s_E , the compact heavy-threshold interval I_E , and the source-qualified comparison class. It proves the existence, response completeness, nonlinear identifiability, and finite-exposure margin of an in-principle physical protocol. It does not assert unrestricted phase tomography, a Lipschitz inverse from a null count to an unsquared amplitude, implementation by a present facility, selection of a unique PCR-5E realization, completion of the parent neutrino problem, or a target-independent lower bound on δ_E .

No converse microscopic-identification claim

Observation of the predicted packet would be consistent with the completed vectorlike-joint carrier architecture but would not identify that microscopic origin uniquely; a richer source theory may reproduce the same record class. Absence of the packet under the complete predeclared protocol is the clean falsifier. Unique microscopic attribution lies outside the theorem.

Contents

Abstract	i
Principal theorem	iii
Scope and interpretation	iii
1 Introduction	1
1.1 From carrier identity to experimental consequence	1
1.2 The charged-current closure relation	1
1.3 Conditions for a prediction lock	1
1.4 Organization	2
2 Foundational setting and scope	2
2.1 Established upstream structure	3
3 Empirical provenance and held-out validation	3
3.1 Channel-level empirical roles	3
4 Observable target and reference theories	4
4.1 No-vectorlike reference classes	4
4.2 Measurement protocols and lawful transformations	5
4.3 Reference-theory normal form	5
5 Observable semantics and detector-record equivalence	5
5.1 Observable trace factors	6
6 Observable atlas and moduli covariance	7
6.1 Atlas strata and lawful overlap	8
7 Physical charged-current tomography and nonlinear confidence geometry	8
7.1 Finite charged-current source and analyzer system	8
7.2 Stratified invariant response coordinates	9
7.3 Uniform response and acceptance	10
7.4 Stratified identifiability and null certification	11
7.5 Finite-sample confidence geometry and separation margin	12
8 Parameter dependence, nuisance quotient, and decoupling	14

8.1	Compact parameter domains and response envelopes	15
8.2	Nuisance transformations and quotient structure	15
9	Cross-child charged-current closure invariant	16
9.1	Source-level identities	16
9.2	Basis-free formulation	17
9.3	Heavy-pole multiplicity as an observable class	18
9.4	Source-matching identity	18
10	No-vectorlike reference responses	18
10.1	Lawful reference presentations	19
11	Detector-level confidence separation	19
11.1	Detector transport as a statistical relation	20
12	Uniformity and nondecoupling across completed sectors	21
12.1	Route-by-route hiding audit	21
13	Target-dependent observables and residual channels	22
13.1	LNV and protector decoration	22
13.2	Residual observable channels	22
13.3	Why no single rate carries the full theorem	23
14	Experimental prediction lock	23
14.1	Necessity and sufficiency	24
14.2	Strength of the result	25
15	Experimental falsifier and measurement strategy	25
15.1	Localization of a failed prediction	26
15.2	Predeclared experimental specification	27
16	Held-out validation and interpretation	27
16.1	Interpretation of validation outcomes	27
17	Consequences for the PCR-8 capstone	28
17.1	Scope of the capstone consequence	28
18	Robustness under same-scope counterexamples	28
19	Conditions for extension or revision	30
20	Conclusion	30
A	Notation and status calculus	31
B	Empirical provenance table	32

C	Physical protocol normal form	32
D	Response-envelope normal forms	33
E	Source and dependency index	34
F	Theorem dependency index	35
G	Supplementary formal representations and reproducibility	37
H	Claim and nonclaim register	38
I	Observable-use grammar	38
I.1	Candidate packet generation	39
J	Decoupling-route analysis	39
J.1	Source-level routes	39
J.2	Presentation routes	39
J.3	Detector routes	39
K	Falsifier localization	40
L	Experimental protocol specification	40
L.1	Measurement objective	40
L.2	Source and response	40
L.3	Count and control model	41
L.4	Finite-exposure margin	41
L.5	Requirements for experimental realization	42
M	Empirical provenance matrix	42
N	Block-unitary derivation of the closure packet	42
N.1	Exact source rotation	43
N.2	Spectral decomposition	43
N.3	Light closure deficit	43
N.4	CLFV residue	43
O	Prediction-status matrix	44
P	Synopsis of principal results	44
	References	46

1 Introduction

1.1 From carrier identity to experimental consequence

A carrier-identity theorem is not yet a prediction theorem. PCR-1 through PCR-5E determine the target-relative neutrino realization space; PCR-6 identifies the terminal bearer structure within each accepted realization; PCR-7 asks whether that structure entails a detector record that survives every remaining lawful freedom. The logical levels are distinct:

$$\text{structural closure} \Rightarrow \text{observable separation} \Rightarrow \text{empirical detection}.$$

A possible response may fail to be invariant, an invariant response may overlap a no-carrier reference, and a separated ideal response may still fail as a prediction if a lawful decoupling or detector-nuisance path reaches the reference closure.

The upstream theorem is unusually specific. On two target-and-comparison-qualified sectors, PCR-6 proves one terminal carrier in every accepted realization, with three light standing sectors and one rank-three degenerate heavy standing sector [Mal26ab]. The bearer core is common to the bounded-accidental and exact-global sectors, while their LNV/protector decoration differs. PCR-5E nevertheless retains continuum many lawful realizations in each completed sector and an unresolved parent complement [Mal26o]. Any observable conclusion must therefore be natural over a groupoid-valued continuum rather than derived from a preferred benchmark.

1.2 The charged-current closure relation

The light and heavy charged-current blocks are

$$N = c_E U, \quad V_H = -s_E U, \quad c_E^2 + s_E^2 = 1,$$

where $s_E > 0$ is fixed by the source-qualified child certificate and U is unitary. The heavy sector contains one threefold-degenerate pole H_λ with λ ranging over a compact interval. Hence

$$NN^\dagger = c_E^2 \mathbf{1}_3, \quad V_H V_H^\dagger = s_E^2 \mathbf{1}_3$$

relate the deficit of the light active block to the inclusive heavy spectral weight. The central observable is therefore not a single event rate but a source-matched charged-current closure packet carrying the active deficit, inclusive heavy weight, and heavy-pole support.

The single-rate alternatives are insufficient. Active normalization alone is calibration-sensitive; heavy production alone is acceptance- and background-sensitive; and a pole alone does not establish that the heavy support closes the missing active weight. The joint packet retains the common source relation and provides a basis-independent comparison with a no-vectorlike reference.

1.3 Conditions for a prediction lock

A target-relative experimental prediction requires more than the source identity. The observable class must be complete before a favorable channel is selected; the packet must be natural across both completed child groupoids; carrier and reference must be evaluated in the same detector quotient; the reference must be specified independently; every source, threshold, parameter, nuisance, and decoupling route must be classified; and the validation record must be withheld until the protocol and falsifier are fixed.

The detector geometry introduces an additional subtlety. The predicted off-diagonal Gram coordinates vanish, so an ordinary count is quadratic rather than linearly sensitive to the corresponding complex amplitude. The protocol therefore reconstructs diagonal coordinates with two-sided intervals and certifies off-diagonal nulls through one-sided upper bounds on invariant squared magnitudes. The finite-exposure analysis uses the resulting nonlinear scaling and proves rank through positive-semidefinite matrix bounds rather than phase reconstruction.

The strongest result is a cross-child prediction lock: the same packet, reference class, physical protocol, confidence rule, and falsifier apply to both completed sectors. The theorem is deliberately narrower than an empirical-discovery claim. It establishes a finite predeclared source-to-record protocol and a positive confidence margin, while leaving actual facility realization and held-out comparison to subsequent experiment.

1.4 Organization

Sections 2 and 3 specify the upstream scope and empirical provenance. Sections 4 and 5 define the observable target, reference theories, semantic observable class, protocol grammar, and detector quotient. Section 6 establishes the observable atlas and its covariance over the completed moduli. Sections 7 and 8 analyze the finite charged-current tomography protocol, nonlinear confidence geometry, nuisance structure, and decoupling atlas. Section 9 derives the common closure packet. Sections 10 to 12 establish reference completeness, detector-level separation, and nondecoupling. Section 13 classifies target-dependent and residual channels, and Section 14 proves the prediction lock together with exact lower statuses. The remaining sections state the falsifier, held-out validation rule, capstone consequences, robustness conditions, and scope of possible revision.

Nonclaim 1.1 (No use of validation data). No measurement from the held-out or future protocol is used to choose $p_\star$, fix δ_E , select the heavy interval, narrow the nuisance set, or establish the separation theorem.

2 Foundational setting and scope

Theorem 2.1 (P7-00 – import, version, target, and no-backflow lock). *Every imported claim used in PCR-7 retains its source target, comparison class, version, and readiness status. The positive domain is exactly*

$$\mathcal{B}_7^{\text{ready}} = \{b_{\text{acc}}, b_{\text{glob}}\}.$$

No D3-1.3 conclusion is transported to D3-1.1, no D4 public-summary result is promoted to the complete cosmological target, and no PCR-7 observable consequence may alter the PCR-5E quotient or the PCR-6 carrier identity.

Proof. The import graph is the accepted chain

$$\text{PCR-5E v1.4.0} \longrightarrow \text{PCR-6 v1.3.0} \longrightarrow \text{PCR-7}.$$

PCR-5E defines the two source-qualified children and preserves the unresolved parent. PCR-6 quantifies only over those children and exports one bearer, the 3 + 1 standing core, the minimal nature packet, and the parent firewall. The D3 successor-branch memorandum supplies the target authority for the 1.3 children but explicitly preserves D3-1.1. The D4 public-summary theorem preserves the missing complete-likelihood gate. Observable projection is downstream of each result, so it cannot be used as a same-scope selector for any upstream quotient. All other imports are typed auxiliary theorems. Therefore the asserted dependency graph has no untagged edge. $\square$

Theorem 2.2 (P7-01 – PCR-6 observable-handoff finality). *The accepted PCR-6 output is complete for forming the branch-indexed PCR-7 observable target. It supplies, for every $b \in \mathcal{B}_7^{\text{ready}}$ and $X \in \text{PhysMod}_{E,VJ}^b$:*

1. *the one-bearer theorem and minimal identity packet;*
2. *the standing spectral core $\{S_1, S_2, S_3, S_H\}$;*
3. *the intrinsic vectorlike source blocks $N = c_E U$ and $V_H = -s_E U$;*
4. *the common-core/target-decoration decomposition;*
5. *lawful groupoid morphisms, continuations, and failure boundaries;*
6. *the explicit parent nonpromotion and projection firewalls.*

No additional carrier-identity premise is required by PCR-7.

Proof. PCR-7 requires only the target-relative carrier packet and the lawful maps needed to transport it into detector records. Each item is contained in the PCR-6 minimal packet and downstream handoff. Observable protocols, detector nuisances, reference responses, and data-status partitions are new PCR-7 objects and are not assumed upstream. Conversely, none of those new objects is needed to state the PCR-6 carrier theorem. The handoff is therefore sufficient and independently formed. $\square$

2.1 Established upstream structure

Object	Accepted status	PCR-7 consequence
Terminal bearer	One inside every accepted X	Defines the source of projected consequences; does not select one X .
Spectral core	Three light sectors plus one heavy-degenerate sector	Supplies light/heavy response blocks without identifying four particles.
Common child core	Same vectorlike carrier core	Permits a cross-child observable search.
Decoration	Accidental and exact-global packets differ	Requires branch-indexed LNV/protector analysis.
PCR-5E quotient D1/D4	Continuum and over-residue Canonical common generator	Requires uniformity over moduli. Mass-sensitive records are correlated but are prior inputs.
D2/D5 exact-global	Obstruction-coupled	Exact-global LNV response is the defined zero subfibre.
D3/D5	Independent on decided rows	Heavy-state and symmetry records cannot be merged into one signal by label.

Warning 2.3 (Parent-scope firewall). The theorem proved here is not

$$\forall X \in \text{NeutrinoUniverse} : \text{PredLock}(X).$$

It is exactly the two-child theorem. An observable in an unresolved PCR-5E parent sector remains outside the completed comparison class.

3 Empirical provenance and held-out validation

Definition 3.1 (Empirical role partition). Let $\mathcal{D}_{\text{all}}$ be the complete record registry used by PCR-1 through PCR-7. Before observable evaluation, partition

$$\mathcal{D}_{\text{all}} = \mathcal{D}_{\text{anchor}} \sqcup \mathcal{D}_{\text{cal}} \sqcup \mathcal{D}_{\text{derive}} \sqcup \mathcal{D}_{\text{hold}} \sqcup \mathcal{D}_{\text{future}}.$$

Anchors form upstream targets; calibration records enter declared response maps; derivation records establish the prediction theorem; held-out records are available but unused before lock; future records do not yet exist.

Theorem 3.2 (P7-02 – empirical-status partition and held-out firewall). *The PCR-7 data registry admits a total role assignment satisfying*

$$(\mathcal{D}_{\text{anchor}} \cup \mathcal{D}_{\text{cal}} \cup \mathcal{D}_{\text{derive}}) \cap \mathcal{D}_{\text{hold}} = \emptyset.$$

The common charged-current spectral-tomography validation record belongs to $\mathcal{D}_{\text{future}}$ at theorem formation. No held-out or future record changes the observable target, protocol, response class, uncertainty threshold, or falsifier.

Proof. Oscillation observations are target anchors. KATRIN, the declared ^{76}Ge response, the D3 heavy/closure/LFU/CLFV response set, the D4 public mass-sum summary, and the D5 structural records are upstream anchors, calibration data, or derivation inputs according to their theorem roles. The electron retained tester and the positive source coefficient s_E are derivation inputs, not predicted validation values. The proposed flavor-complete active/heavy closure protocol is neither used by those upstream theorems nor tuned against an observed packet; its complete record is therefore assigned to $\mathcal{D}_{\text{future}}$. The formal registry is total by provenance and theorem-use fields, and the role categories are disjoint by definition. Hence the firewall holds. $\square$

3.1 Channel-level empirical roles

Channel	Upstream role	PCR-7 authority
Oscillation boundary and phase response	Target anchor	Not an independent prediction; only new derived relations may be held out.
KATRIN endpoint ^{76}Ge LNV response	D1 target/calibration D2 target/calibration	Input at the same response level. Same-isotope response is input; another protocol may supply a branch test.
D3 heavy/active/LFU/CLFV records	Target and solved-region inputs	Existing charts do not validate themselves; the new joint tomography is a derived protocol.
D4 public mass-sum summary	Target input	Not a prediction; complete cosmology remains NotReady.
D5 action/protector	Structural input	May imply exact branch selection rules only through a new protocol.
Flavor-complete active/heavy tomography	No upstream use	Future pre-registered validation protocol.

Nonclaim 3.3 (No empirical confirmation). The status `FrozenPredictionAwaitingHeldOutData` is not evidence that the predicted packet has been observed. It states only that the protocol and response class are fixed before future comparison.

4 Observable target and reference theories

Definition 4.1 (Branch-indexed observable target). For $b \in \mathcal{B}_7^{\text{ready}}$, define

$$T_{\text{obs}}^{\nu,b} = (T_{\text{nat}}^{\nu,b}, \text{Prot}_b, \text{Rec}_b, \Theta_b, \text{Nuis}_b, \text{Ref}_b, \text{DataStatus}_b, \text{Cont}_b, \text{Fail}_b).$$

The target asks which detector-record consequences of the accepted carrier are invariant, reference-separated, and nondecoupling. It contains no preferred experiment, desired signal size, or validation outcome.

Theorem 4.2 (P7-03 – observable target formation). *The objects $T_{\text{obs}}^{\nu,b}$ can be formed before a candidate observable is selected. They preserve the PCR-6 minimal carrier packet and child tag, admit scalar/vector/higher-arity protocols, and separate target updates from lawful detector continuation.*

Proof. The source objects are the PCR-6 minimal nature packet, its lawful arrows, and its projection firewall. Protocol sorts are generated by target-visible boundary, process, spectral, matching, and correlation uses, not by a chosen experiment. Record sorts, nuisance objects, and reference objects are declared independently. A change of source ontology, ready child, detector principle, or record semantics is an update; changes inside a declared calibration/response object are lawful continuations. No endpoint-bearing record appears in target formation. $\square$

4.1 No-vectorlike reference classes

Definition 4.3 (No-vectorlike-carrier references). The primary reference family contains two independently frozen objects.

1. R_0^{U3} preserves the known charged-current boundary and a unitary three-light response, with $N_0 = U_0$ and $V_{H,0} = 0$.
2. R_0^{SM0} preserves the minimal massless-neutrino Standard Model boundary and detector substrate, again with no heavy pole and no active deficit.

Neither is a survivor of the PCR-5E child target; both are comparison objects formed before $O_\star$ is evaluated.

Theorem 4.4 (P7-10 – reference target formation). *The reference family is formed without inspecting the carrier response. It preserves the detector, backgrounds, calibration conventions, light charged-current boundary, and record quotient while forbidding compensating vectorlike operators or post-hoc nonunitarity.*

Proof. Start from the known-substrate boundary objects already present before the vectorlike extension. Apply the no-repair deletion of the vectorlike heavy source and restore only the independently defined unitary/minimal light reference law; no parameter is fitted to the carrier prediction. Detector and nuisance objects are copied by identity. A compensating heavy pole, active deficit, or new operator would change the reference source and is therefore an explicit update. The construction is independent of $O_\star$. $\square$

Remark 4.5. The phrase “no carrier” is reference-relative. The comparator preserves the known light weak boundary needed to define the same protocol, but omits the terminal vectorlike-joint completion whose observable consequence is under test.

4.2 Measurement protocols and lawful transformations

Definition 4.6 (Protocol category). An object of Prot_b is a finite typed intervention/detector diagram

$$p = (I_p, D_p, C_p, Q_p, F_p)$$

consisting of intervention slots, detector transport, calibration/control data, a lawful record quotient, and a total applicability/failure map. A morphism $p \rightarrow p'$ is a standing-preserving refinement that leaves the physical response semantics fixed while adding resolution, control information, or a lawful record map.

Lemma 4.7 (Protocol refinement monotonicity). *If $p \rightarrow p'$ is a lawful refinement and the p -record classes are separated after transport, then the pulled-back p' classes remain separated unless the refinement changes the source, reference, or record equivalence.*

Proof. A lawful refinement factors through the original record map and preserves the separator. If two refined classes met, their images would meet in the original quotient. A refinement that adds a compensating source term or changes which records are identified is a target update rather than a morphism in Prot_b . $\square$

4.3 Reference-theory normal form

Proposition 4.8 (Reference normal-form exhaustion). *Every same-protocol no-vectorlike reference presentation is equivalent to a member of the family generated by R_0^{U3} and R_0^{SM0} together with lawful detector/background/calibration presentations. A reference containing nonzero heavy charged-current weight, a heavy pole, or intrinsic active nonunitarity lies outside the reference class.*

Proof. The independently fixed reference target preserves the known light boundary and detector but deletes the vectorlike extension without repair. The remaining source choice is either the unitary three-light comparator or the minimal massless boundary comparator, up to lawful known-substrate enrichment. Detector/background changes are presentations if they preserve the complete reference trace. Nonzero heavy weight or active nonunitarity changes the source-facing response and cannot be introduced by presentation. Thus the normal form is exhaustive. $\square$

5 Observable semantics and detector-record equivalence

Definition 5.1 (Semantic observable use). A semantic observable use of a child realization is a target-authorized operation that transports a source/physical response through a declared intervention and detector system to a reusable record while preserving child identity, protocol identity, applicability, and failure semantics.

Theorem 5.2 (P7-04 – semantic observable-use class). *For each $b \in \mathcal{B}_7^{\text{ready}}$ there is an extensional trace*

$$\tau_{\text{obs}}^b : \text{UseObs}_b \longrightarrow \text{TraceObs}_b$$

whose factors are boundary insertion, process response, spectral support, matching/continuation, detector transport, nuisance action, record status, and failure/update data. Equality of complete traces is semantic observable equivalence.

Proof. PCR-2 supplies target-facing boundary and process uses; PCR-5C supplies lawful restriction and continuation; PCR-5E/PCR-6 supply the child source, one-apex physical object, and carrier projection. Any observable

use changes either an intervention/boundary insertion, a source response, detector transport, record equivalence, or status. These are precisely the trace factors. If all factors agree, no target-admitted observable continuation distinguishes the uses; if one differs, the corresponding use witnesses the difference. Thus the trace is extensional. $\square$

Definition 5.3 (Protocol grammar). The protocol source theory is generated by finite typed compositions of:

Prepare, Interact, Propagate, Scan, Tag, Reconstruct, Normalize, Correlate, Calibrate, Transport, Quotient.

It permits scalar records, matrix tomography, spectral measures, shape/ratio records, and finite higher-arity packets.

Theorem 5.4 (P7-05 – protocol/source grammar completeness). *Every finitely presented semantic observable use in the declared PCR-7 comparison class has a protocol normal form generated by the preceding operations, and every generated normal form has a coherent semantic interpretation. The grammar is fixed before channel ranking.*

Proof. Decompose a semantic use along its finite causal/intervention graph. Source and boundary insertions give preparation and interaction nodes; target response gives propagation/process nodes; spectral and threshold queries give scans; detector maps give tags/reconstruction; nuisance handling gives calibration/normalization; multi-channel use gives correlation; and lawful equivalence gives quotienting. The finite-presentation assumption inherited from PCR-3/PCR-5B makes the decomposition finite. Conversely, each generator is a typed target-admitted operation and finite composition preserves coherence. Since the grammar is derived from the semantic trace before candidates are scored, survivor backfitting is excluded. $\square$

Definition 5.5 (Detector-record quotient). For protocol p , raw records are equivalent when every target-admitted detector use agrees after all lawful calibration, binning, basis, gauge, phase, chart, and analysis-coordinate changes. The quotient is denoted QRec_p .

Theorem 5.6 (P7-06 – detector-record quotient). *The detector-record relation is the unique same-protocol standing quotient compatible with trace soundness, trace completeness, lawful nuisance transport, and reuse. In particular, it does not collapse a heavy-pole support class into a smooth-background class or a nonzero closure matrix into zero.*

Proof. The proof is the SCIC stable-reference argument applied at detector scope. A finer quotient preserves an unwitnessed presentation distinction; a coarser quotient merges records distinguished by a target-admitted detector continuation. Pole support, matrix rank, and the calibrated active/heavy closure relation are retained by the semantic trace, so they survive. Gauge/basis/phase labels that do not change the trace collapse. $\square$

5.1 Observable trace factors

For later use, write the complete trace as

$$\tau_{\text{obs}}^b(u) = (\tau_{\partial}, \tau_{\text{proc}}, \tau_{\text{spec}}, \tau_{\text{match}}, \tau_{\text{det}}, \tau_{\text{nuis}}, \tau_{\text{rec}}, \tau_{\text{fail}}).$$

The factors respectively record boundary preparation/detection, process response, spectral support, threshold/-matching continuation, detector transport, nuisance orbit, quotient record, and applicability/failure status.

Lemma 5.7 (Trace-factor independence). *No trace factor can be deleted uniformly from the complete semantic observable class.*

Proof. Deleting τ_{∂} loses flavor/boundary typing; deleting τ_{proc} identifies different physical responses; deleting τ_{spec} merges pole and continuum support; deleting τ_{match} loses target-preserving threshold continuation; deleting τ_{det} leaves no record; deleting τ_{nuis} confuses nominal and complete envelopes; deleting τ_{rec} removes stable reference; and deleting τ_{fail} launders inapplicability into a physical null. Each deletion admits a target-relevant counterexample from the D-series source stack. $\square$

Proposition 5.8 (Higher-arity closure of the protocol grammar). *Finite correlation packets introduce no new primitive protocol generator beyond Correlate and the primitive trace factors. Their standing content is the physical isomorphism class of the common-source incidence diagram.*

Proof. A finite packet is a finite family of marginal protocol diagrams together with shared source, calibration, nuisance, and record relations. The marginals are generated by P7-05; the shared relations are expressed by Correlate and typed pullback over the common source. Any additional primitive would either be a new detector principle/source operation or a presentation of the same incidence diagram. $\square$

6 Observable atlas and moduli covariance

Theorem 6.1 (P7-07 – observable map theorem). *Every protocol normal form p induces a total fail-closed map*

$$O_{b,p} : \text{PhysMod}_{E,VJ}^b \times \Theta_b \times \text{Nuis}_p \longrightarrow \text{QRec}_p \cup \{\text{NotReady}, \text{TargetUpdate}\},$$

with exact applicability and failure status.

Proof. Evaluate the source/physical response on the accepted one-apex object, transport it through the protocol, apply the nuisance action, and descend to the detector quotient. Undefined source support, inapplicable kinematics, incomplete detector response, and target updates receive distinct typed outcomes. Congruence follows from source/physical and record equivalence. $\square$

Theorem 6.2 (P7-08 – moduli naturality). *For every lawful arrow $f : X \rightarrow Y$ in a ready child groupoid and every compatible protocol/nuisance transport,*

$$O_{b,p}(Y) \simeq O_{b,p}(X)$$

in QRec_p whenever the observable is classified as carrier-core invariant. Parameter-dependent observables transport naturally to their branch envelopes.

Proof. Lawful arrows preserve the PCR-6 carrier core, child target, source equations, boundary, and continuation signature. The protocol functor is built from target-admitted operations and is therefore natural under those arrows. Quotient descent removes presentation changes. Coordinates that vary by lawful source/enriched parameters map to the corresponding envelope rather than a point; the envelope assignment is functorial. $\square$

Theorem 6.3 (P7-09 – observable atlas exhaustion). *The complete observable atlas decomposes into the following target-generated classes:*

1. *oscillation/boundary response;*
2. *direct kinematic endpoint and absolute-scale response;*
3. *LVN/conjugation response;*
4. *heavy direct-state, active-closure, universality, and CLFV response;*
5. *cosmological population/background response;*
6. *symmetry/charge/protector response;*
7. *finite cross-channel response packets generated by shared carrier incidences.*

Every semantic observable use factors through one class or a finite higher-arity combination. The typed complement consists only of protocol/source updates or incomplete detector maps.

Proof. The PCR-5E standing-coordinate incidence relation exhausts the admitted D1–D5 target faces, while P7-04/P7-05 exhaust observable transport and correlation. A new same-scope use must therefore enter one marginal face or a finite relation among them. If it introduces a new target observable, source operator, detector principle, or background class, it is an update. If its transport is incomplete, it is NotReady. No further same-scope class remains. $\square$

Theorem 6.4 (P7-R1 – higher-arity cross-channel observable theorem). *The active-light deficit, inclusive heavy spectral weight, and heavy-pole support define a genuine higher-arity observable packet. Their common source incidence is standing-visible and cannot be replaced, at the complete protocol, by any one marginal coordinate alone.*

Proof. The active deficit alone is vulnerable to overall charged-current normalization; heavy spectral weight alone is vulnerable to protocol acceptance/background representation; pole support alone does not connect the pole to the active closure deficit. The common source identities $N = c_E U$ and $V_H = -s_E U$ imply the exact relation $\Delta_{\text{act}} = \Omega_H$, and the same heavy block supplies the pole class. Deleting any leg removes one target-authorized correlation use. Hence the packet is irreducibly higher-arity at the complete detector target. $\square$

6.1 Atlas strata and lawful overlap

For each protocol type a , let $\mathcal{U}_a^b$ be the groupoid of source/parameter/nuisance points on which that protocol is applicable. Pairwise and higher overlaps are homotopy pullbacks over the common detector-record core. Codescent yields the global observable groupoid

$$\mathcal{A}_{\text{obs}}^b \simeq \text{codesc}\{\mathcal{U}_a^b\}_{a \in I_{\text{obs}}}.$$

Proposition 6.5 (No merger of separated record classes). *Two observable strata whose complete detector traces differ in pole support, matrix rank, or calibrated closure residual cannot merge under atlas codescent.*

Proof. Overlap arrows are lawful protocol/source/record isomorphisms. They preserve every target-admitted trace coordinate. Pole support, matrix rank, and closure residual are retained coordinates of QRec_p . Therefore a codescent identification between different values would violate trace soundness. $\square$

Remark 6.6. The observable atlas is generally larger than the prediction atlas. Most observable strata are inputs, conditional responses, or overlapping envelopes. Prediction status is assigned only after reference and decoupling analysis.

7 Physical charged-current tomography and nonlinear confidence geometry

The structural packet becomes an experimental prediction only after transport through a physical source-to-record system with fixed applicability, response, reconstruction, nuisance structure, and uncertainty rule, all specified before held-out signal data are examined. This section specifies and analyzes such a system on the two completed child scopes. The relevant record geometry is stratified: diagonal Gram coordinates are reconstructed with ordinary two-sided errors, whereas predicted off-diagonal nulls are represented by squared-modulus coordinates and certified with one-sided upper bounds. No differentiable or Lipschitz inverse from a rate at zero amplitude to the unsquared amplitude is assumed.

7.1 Finite charged-current source and analyzer system

Write the compact heavy interval as

$$H(I_E) = [H_-, H_+].$$

Choose a compact interval of timelike charged-current parent masses

$$J_Q = [Q_-, Q_+], \quad Q_- > H_+ + m_\tau + \Delta_Q$$

for some fixed $\Delta_Q > 0$. A tagged parent event is a source-authorized timelike charged-current insertion $W^*(Q)$, $Q \in J_Q$, together with control records for the parent flux and kinematics. The insertion may be realized by any physical production process that supplies the declared charged-current parent; the theorem depends on the tagged instrument, not on a preferred accelerator. Charged-current production and prompt or displaced decay/scattering channels of heavy neutral leptons are established physical process classes [Atr+09; Aba+22; ATL25a; ATL25b].

For each charged-lepton boundary flavor $\alpha \in \{e, \mu, \tau\}$, the light and heavy charged-current couplings are

$$\mathcal{L}_{\text{CC}} \supset -\frac{g}{\sqrt{2}} W_\mu^{*+} \bar{\ell}_\alpha \gamma^\mu P_L (N_{\alpha i} \nu_i + (V_H)_{\alpha a} N_a) + \text{h.c.},$$

with

$$N = c_E U, \quad V_H = -s_E U.$$

The coefficient $s_E > 0$, the light phase-frequency data, and the compact heavy interval are fixed by the accepted child certificate before protocol construction. Let $\mathcal{K}_\star$ denote the finite family of compact source, propagation, analyzer, mass-response, and control cells selected below.

Construction 7.1 (Finite dual-arm charged-current spectral-tomography protocol). The protocol $p_\star^{\text{fin}}$ is the finite union of the following instruments.

1. **Parent and detector controls.** Tagged parent events, charged-lepton flavor controls, fiducial and timing controls, mass-response controls, independent efficiency samples, and disjoint background sidebands determine the finite calibration and nuisance coordinates.
2. **Light arm.** For every source flavor α , the tagged current produces the light state coherently with ℓ_α . For every analyzer flavor β , charged-current detection is performed at seven predeclared values

$$x_j := \frac{L_j}{2E_j}, \quad j = 1, \dots, 7,$$

chosen so that the seven functions

$$1, \cos(\Delta m_{21}^2 x), \sin(\Delta m_{21}^2 x), \cos(\Delta m_{31}^2 x), \sin(\Delta m_{31}^2 x), \cos(\Delta m_{32}^2 x), \sin(\Delta m_{32}^2 x)$$

have an invertible evaluation matrix. The calibrated light records determine the diagonal entries of $NN^\dagger$ and the squared moduli of its off-diagonal entries at the target record quotient.

3. **Heavy production arm.** For every source flavor α , inclusive tagged production

$$W^*(Q) \longrightarrow \ell_\alpha + N_a$$

is recorded over a finite cover of $J_Q \times H(I_E)$. Summing over the unresolved exactly degenerate heavy states gives a rate proportional to $(\Omega_H)_{\alpha\alpha}$.

4. **Heavy analyzer arm.** The produced heavy state is followed by every applicable flavor-tagged charged-current analyzer: a visible decay

$$N_a \longrightarrow \ell_\beta + X_{CC}$$

or a charged-current scattering cell

$$N_a + T_\beta \longrightarrow \ell_\beta + X_\beta.$$

Analyzer energies are chosen above the relevant charged-lepton thresholds. Exact heavy degeneracy makes the unresolved two-current amplitude proportional to $(\Omega_H)_{\beta\alpha}$. The calibrated count probability is therefore linear in the invariant quadratic coordinate $|(\Omega_H)_{\beta\alpha}|^2$. Predicted off-diagonal nulls are tested by one-sided upper confidence bounds on this squared-modulus coordinate; no linear-amplitude derivative at the null is assumed. A kinematically closed cell retains exact N/A status and is replaced only by another source-authorized cell from the finite cover.

5. **Pole support.** Missing-mass or visible invariant-mass reconstruction is performed on the same finite heavy cells. The target record is the existence of one rank-three degenerate pole class in $H(I_E)$, not an exact real-valued mass estimate.
6. **Common carrier/reference quotient.** Carrier and no-vectorlike reference records use the same parent instrument, analyzers, calibration, backgrounds, response maps, and quotient. On the reference, heavy signal coordinates are physical zero-amplitude records rather than undefined entries.
7. **Blind split.** Calibration, response identification, and sideband estimation use disjoint control samples. The final signal-region packet is held out until the source cells, analyzer cells, exposure allocation, response model, nuisance domain, reconstruction relation, confidence level, and falsifier are frozen.

The protocol record is a finite vector of normalized light probabilities, inclusive heavy-production probabilities, correlated heavy-analyzer probabilities, pole-support bins, control frequencies, sideband frequencies, and their covariance data.

7.2 Stratified invariant response coordinates

For a light source/analyzer pair define

$$\mathcal{A}_{\beta\alpha}^L(x) = K_{\beta\alpha}^L(x) \sum_{i=1}^3 N_{\beta i} N_{\alpha i}^* e^{-im_i^2 x}.$$

Its normalized probability is a real trigonometric polynomial with one constant and three complex pair coefficients. The seven functions in [Theorem 7.1](#) are linearly independent, so generic predeclared points x_j give an invertible seven-by-seven sampling matrix. Evaluating the reconstructed polynomial at $x = 0$ gives

$$q_{\alpha\beta}^L := |(NN^\dagger)_{\beta\alpha}|^2.$$

For $\alpha = \beta$, positivity and the fixed carrier/reference lower bound determine

$$d_\alpha^L := (NN^\dagger)_{\alpha\alpha}.$$

The corresponding active-deficit coordinates are

$$d_\alpha^\Delta := 1 - d_\alpha^L, \quad q_{\alpha\beta}^\Delta := q_{\alpha\beta}^L \quad (\alpha < \beta).$$

For a heavy source/analyzer pair, exact degeneracy gives a common propagator D_H and common kinematic response on each cell. Hence

$$\mathcal{A}_{\beta\alpha}^H(q^2) = K_{\beta\alpha}^H(q^2)D_H(q^2)(\Omega_H)_{\beta\alpha}.$$

Inclusive production satisfies

$$\Gamma_\alpha^H(Q, H) = \kappa_\alpha^H(Q, H)d_\alpha^H, \quad d_\alpha^H := (\Omega_H)_{\alpha\alpha},$$

while the calibrated correlated analyzer probability has the form

$$p_{\beta\alpha}^H = \kappa_{\beta\alpha}^H(H, \eta)q_{\alpha\beta}^H + \beta_{\beta\alpha}^H(\eta), \quad q_{\alpha\beta}^H := |(\Omega_H)_{\beta\alpha}|^2, \quad \alpha < \beta.$$

The invariant record coordinate is therefore

$$\Xi_{\text{strat}} = \left((d_\alpha^\Delta)_\alpha, (d_\alpha^H)_\alpha, (q_{\alpha\beta}^\Delta)_{\alpha < \beta}, (q_{\alpha\beta}^H)_{\alpha < \beta}, \text{Pole}_H \right).$$

It is a quotient coordinate: phases of nonzero off-diagonal entries are not reconstructed unless separately activated. On the carrier and reference comparison classes used here, all off-diagonal coordinates vanish, so Ξ_{strat} is complete for the target comparison without claiming unrestricted matrix tomography.

7.3 Uniform response and acceptance

Lemma 7.2 (Uniform source-to-record kernels). *For every ready child and every $H \in H(I_E)$, the protocol contains a finite set of source-authorized cells whose required light and heavy response kernels are continuous on compact interior subcells. There are constants*

$$a_\star > 0, \quad \kappa_{d,\star} > 0, \quad \kappa_{q,\star} > 0$$

such that every nonzero diagonal packet coordinate has acceptance at least $a_\star$, every retained diagonal response coefficient is at least $\kappa_{d,\star}$, and every retained squared-modulus response coefficient is at least $\kappa_{q,\star}$. Predicted null off-diagonal coordinates remain defined signal coordinates and are tested by one-sided bounds on q^Δ and q^H .

Proof. The inequality $Q_- > H_+ + m_\tau + \Delta_Q$ places every heavy-production flavor inside an open phase-space region. Because $s_E > 0$, each inclusive source row has positive total heavy weight $(\Omega_H)_{\alpha\alpha} = \delta_E$. The light boundary channels are already source-authorized by the accepted oscillation interface. For each heavy analyzer flavor, at least one decay or scattering channel is open after the analyzer energy is chosen above threshold; the coefficient multiplying $|(\Omega_H)_{\beta\alpha}|^2$ is continuous and positive on an interior cell. The fixed positive mixing gives a positive total heavy width throughout the compact interval, and the source/fiducial geometry can be chosen with positive decay or scattering acceptance. The open positive-response cells cover the compact heavy interval. After local physical positivity is established, compactness gives a finite subcover and positive minima $a_\star, \kappa_{d,\star}, \kappa_{q,\star}$. At a predicted null the derivative with respect to the unsquared amplitude may vanish, but the record remains linear with positive coefficient in the invariant coordinate $q = |\Omega_{\alpha\beta}|^2$; this is the coordinate used by the protocol. $\square$

Theorem 7.3 (P7-S1 – source-authorized CCST physical-protocol realization). *There exists a finite charged-current spectral-tomography protocol $p_\star^{\text{fin}}$ formed from source-authorized preparation, light and heavy production, propagation, decay or scattering, flavor tagging, mass reconstruction, calibration, sideband, and quotient operations, with the following properties.*

1. *Every signal and control record is the image of a source-coherent process on the same ready-child physical object.*
2. *The complete heavy interval is covered by finitely many response cells with one derived acceptance lower bound $a_\star > 0$.*

3. The same parent, calibration, background, reconstruction, and quotient are defined on the no-vectorlike references; their heavy signal coordinates are physical zeros rather than undefined records.
4. Calibration, acceptance, mass response, backgrounds, and reconstruction form a finite independently controlled nuisance object.
5. The complete record determines the stratified coordinate Ξ_{strat} through positive diagonal and squared-modulus response coefficients and a separately calibrated pole-support record.
6. Source geometry, response cells, nuisance class, reconstruction relation, and falsifier are fixed before held-out signal-region records are inspected.

Thus physical response formation, uniform acceptance, carrier/reference comparability, and a finite target-adapted record quotient are established.

Proof. Physical formation and positive response follow from [Theorems 7.1 and 7.2](#). The exact light and heavy response factorizations above identify the invariant diagonal and squared-modulus coordinates without introducing a basis-dependent heavy-column list. Carrier and reference use the same raw instrument and nuisance coordinates, so reference zeros remain defined signal coordinates. All design choices depend only on the frozen child certificate and compact source/response domains and precede held-out signal data. $\square$

7.4 Stratified identifiability and null certification

Let $r(H, \eta)$ denote the finite vector of ideal normalized record probabilities, where η contains the finite calibration, efficiency, resolution, and background coordinates. After the predeclared light-sampling transform and common control calibration, the retained record blocks factor through

$$d := (d^\Delta, d^H), \quad q := (q^\Delta, q^H), \quad \text{Pole}_H.$$

The diagonal and squared-modulus blocks have separate condition constants. No square-root inversion is used at $q = 0$.

Theorem 7.4 (P7-S1A – stratified quadratic-record identifiability and one-sided null certification). *On the frozen carrier/reference comparison family there are finite constants*

$$L_d < \infty, \quad L_q < \infty$$

and a compact nuisance domain $E_\star$ such that the calibrated finite record determines the stratified quotient coordinate

$$\Xi_{\text{strat}} = (d, q, \text{Pole}_H)$$

with the following properties.

1. Diagonal-coordinate perturbations obey a uniform two-sided bound

$$\|\widehat{d} - d\|_2 \leq L_d \|\widehat{r} - r\|_2.$$

2. Squared-modulus perturbations obey a uniform bound

$$\|\widehat{q} - q\|_\infty \leq L_q \|\widehat{r} - r\|_\infty.$$

For a predicted null $q_j = 0$, the protocol reports a one-sided upper confidence bound q_j^U ; it does not infer $|\Omega_j|$ through a Lipschitz square-root map.

3. The pole-support record is identifiable on the declared finite mass-response partition.
4. Ξ_{strat} is injective on the union of the two carrier response classes and the two no-vectorlike reference classes. Any phase directions not determined by (d, q) are quotient-trivial for this comparison and are not claimed to be reconstructed.

Proof. The seven-point light design has a fixed invertible sampling matrix, so the trigonometric coefficients and the values $q_{\alpha\beta}^L = |(NN^\dagger)_{\beta\alpha}|^2$ are finite linear functionals of the calibrated record. On the carrier/reference family the light diagonal values are bounded away from zero, so the positive square-root that recovers d_α^L is Lipschitz

on that compact interval. Heavy inclusive production is linear in d_α^H , and each correlated heavy analyzer record is linear in $q_{\alpha\beta}^H$ with coefficient bounded below by [Theorem 7.2](#). Independent controls fix the finite nuisance coefficients before signal inversion. These facts give finite constants L_d, L_q . At $q = 0$ the squared-modulus coordinate remains linearly identifiable even though the map from q to $\sqrt{q}$ is not Lipschitz. Finally, the carrier classes have $d = (\delta_E \mathbf{1}_3, \delta_E \mathbf{1}_3)$ and a heavy pole, whereas the references have $d = 0$ and no heavy pole, so the stratified quotient is injective on the declared comparison family. $\square$

Theorem 7.5 (P7-S1B – positive-rank certification from diagonal lower bounds and quadratic null bounds). *Suppose a simultaneous confidence relation satisfies, for every flavor α and every $\alpha < \beta$,*

$$|\widehat{d}_\alpha^\Delta - \delta_E| < \frac{\delta_E}{8}, \quad |\widehat{d}_\alpha^H - \delta_E| < \frac{\delta_E}{8},$$

and

$$(q_{\alpha\beta}^\Delta)^U < \left(\frac{\delta_E}{16}\right)^2, \quad (q_{\alpha\beta}^H)^U < \left(\frac{\delta_E}{16}\right)^2.$$

Then every Hermitian positive-semidefinite pair (Δ, Ω) consistent with the confidence relation obeys

$$\lambda_{\min}(\Delta) \geq \frac{3\delta_E}{4} > 0, \quad \lambda_{\min}(\Omega) \geq \frac{3\delta_E}{4} > 0,$$

and hence

$$\text{rank } \Delta = \text{rank } \Omega = 3.$$

The corresponding reference confidence relation with all six diagonal coordinates within $\delta_E/8$ of zero is disjoint from the carrier diagonal confidence relation. Moreover, for either carrier matrix G ,

$$\|G - \delta_E \mathbf{1}_3\|_F^2 < 3 \left(\frac{\delta_E}{8}\right)^2 + 6 \left(\frac{\delta_E}{16}\right)^2 = \frac{9}{128} \delta_E^2.$$

Hence the pair (Δ, Ω) lies within product-space radius

$$\varepsilon_\star^{\text{mat}} := \frac{3}{8} \delta_E$$

of $(\delta_E \mathbf{1}_3, \delta_E \mathbf{1}_3)$; the reference pair obeys the same radius bound about zero. The pole-support classes are separately disjoint.

Proof. For either matrix, every diagonal entry is at least $7\delta_E/8$. Each off-diagonal magnitude is less than $\delta_E/16$, so the sum of the two off-diagonal magnitudes in any row is less than $\delta_E/8$. Gershgorin's theorem therefore gives

$$\lambda_{\min} \geq \frac{7\delta_E}{8} - \frac{\delta_E}{8} = \frac{3\delta_E}{4} > 0.$$

This argument is independent of the unmeasured off-diagonal phases. The Frobenius estimate follows from

$$\|G - \delta_E \mathbf{1}_3\|_F^2 = \sum_\alpha |G_{\alpha\alpha} - \delta_E|^2 + 2 \sum_{\alpha < \beta} |G_{\alpha\beta}|^2.$$

The displayed coordinate bounds give $9\delta_E^2/128$ for each matrix and therefore $9\delta_E^2/64$ for the matrix pair. For the reference, every diagonal coordinate is at most $\delta_E/8$, whereas every carrier diagonal is at least $7\delta_E/8$; the intervals are disjoint. Rank three with a pole in $H(I_E)$ is also disjoint from rank zero/no pole. $\square$

7.5 Finite-sample confidence geometry and separation margin

Let M_d be the finite number of retained diagonal coordinates and M_q the finite number of retained squared-modulus null coordinates. Set

$$\eta_d := \frac{\delta_E}{8}, \quad \eta_q := \left(\frac{\delta_E}{16}\right)^2, \quad 0 < \alpha_\star < 1.$$

All signal and control records may be represented as normalized frequencies of finitely many disjoint tagged trials.

Lemma 7.6 (Stratified finite-sample confidence certificate). *Finite signal, calibration, background, and response-control sample sizes exist such that, with probability at least $1 - \alpha_\star$, every diagonal error is below η_d , every predicted-null squared-modulus coordinate has one-sided upper bound below η_q , and the pole-support class is correctly localized. Conservative sufficient trial counts have the form*

$$n_d \geq \frac{2L_d^2 M_d}{\eta_d^2} \log \left(\frac{8(M_d + M_q)}{\alpha_\star} \right),$$

$$n_q \geq \frac{2L_q^2 M_q}{\eta_q^2} \log \left(\frac{8(M_d + M_q)}{\alpha_\star} \right).$$

Consequently

$$n_d = O \left(\delta_E^{-2} \log \frac{1}{\alpha_\star} \right), \quad n_q = O \left(\delta_E^{-4} \log \frac{1}{\alpha_\star} \right)$$

for fixed response coefficients and finite record dimension.

Proof. Each tagged trial has a finite categorical outcome; normalized cell frequencies are bounded averages. Hoeffding's inequality and a union bound give simultaneous raw-frequency bounds [Hoe63]. P7-S1A transports diagonal and squared-modulus record errors separately by the finite constants L_d, L_q . Taking the raw diagonal tolerance η_d/L_d and the raw quadratic tolerance η_q/L_q gives the displayed counts. Calibration, background, and response-control blocks are finite and receive fixed shares of the same error and confidence budgets. The different δ_E -scalings arise because an amplitude null of size $\delta_E/16$ is certified through a squared-modulus rate of order δ_E^2 . Model failure or an uncalibrated response is not statistical fluctuation; it is routed to `NotReadyprotocol` or update. $\square$

Theorem 7.7 (P7-S2A – nonlinear finite-exposure certified-margin theorem). *For every fixed simultaneous confidence level $1 - \alpha_\star$, finite predeclared signal, calibration, background, and response-control exposures can be chosen so that, on one joint coverage event,*

1. the six diagonal coordinates have Euclidean confidence radius

$$\varepsilon_\star^{\text{diag}} \leq \sqrt{6} \eta_d = \frac{\sqrt{6}}{8} \delta_E < \frac{\sqrt{6}}{2} \delta_E;$$

2. every predicted-null squared-modulus coordinate has one-sided upper bound below η_q ;
3. P7-S1B certifies rank three for both carrier matrices and excludes the reference diagonal class;
4. the heavy-pole confidence class is contained in $H(I_E)$ and is disjoint from `NoHeavyPole`;
5. the complete continuous matrix packet is contained in a product-space ball of radius

$$\varepsilon_\star := \varepsilon_\star^{\text{mat}} \leq \frac{3}{8} \delta_E.$$

The carrier/reference confidence regions therefore satisfy

$$\text{Gap}^{\text{conf}} \geq \sqrt{6} \delta_E - 2\varepsilon_\star \geq \left(\sqrt{6} - \frac{3}{4} \right) \delta_E > 0.$$

Proof. Use Theorem 7.6 with $\eta_d = \delta_E/8$ and $\eta_q = (\delta_E/16)^2$, distributing the confidence budget over signal and control blocks. The six diagonal coordinates contribute Euclidean radius at most $\sqrt{6} \eta_d$. For each Hermitian matrix, the three diagonal errors contribute at most $3\delta_E^2/64$ to the squared Frobenius norm and the three off-diagonal magnitude bounds contribute at most $3\delta_E^2/128$; for the pair (Δ, Ω) the resulting product radius is $3\delta_E/8$. The carrier and reference ideal packet vectors differ by product Frobenius norm $\sqrt{6} \delta_E$, so the triangle inequality gives both displayed positive gaps. The finite mass-response partition supplies the pole statement. All exposures are finite because $\delta_E > 0$, the response coefficients are bounded below, and the record alphabet is finite. $\square$

Theorem 7.8 (P7-S2 – finite-protocol completeness and pre-data freeze). *The physical protocol of P7-S1, the stratified identifiability relation of P7-S1A, the rank certificate of P7-S1B, and the finite-exposure theorem P7-S2A form a complete finite protocol certificate*

$$\text{ProtocolComplete}_{P_\star^{\text{fin}}}.$$

The source/analyzer cells, exposure allocation, calibration and background model, diagonal and quadratic confidence rules, pole reconstruction, record quotient, confidence level, and falsifier can all be frozen before held-out signal records are inspected.

Proof. P7-S1 supplies physical formation, common carrier/reference response, uniform acceptance, and the finite record quotient. P7-S1A proves target-relative injectivity in the stratified coordinates and provides one-sided certification at nulls. P7-S1B turns those record bounds into a rank-three matrix certificate without a false linear-amplitude inverse. P7-S2A supplies finite exposures and a positive carrier/reference confidence margin. Every object is a function of the frozen child certificate, physical response, control geometry, and predeclared confidence level, so the complete certificate can be fixed before signal data. $\square$

Protocol authority

P7-S1, P7-S1A, P7-S1B, P7-S2A, and P7-S2 prove existence of a finite physical protocol at the exact source-qualified child scope. They do not claim unrestricted phase tomography, a Lipschitz inverse from a null rate to an unsquared amplitude, or implementation by a present facility. A future implementation must instantiate the theorem's finite fields without changing the frozen packet or falsifier.

8 Parameter dependence, nuisance quotient, and decoupling

Definition 8.1 (Lawful parameter object). For fixed $X \in \text{PhysMod}_{E,VJ}^b$, write

$$\Theta_b^{\text{law}}(X) = \Theta_{\text{fixed}} \times \Theta_{\text{source}} \times \Theta_{\text{enriched}} \times \Theta_{\text{protocol}}.$$

At the ready children, s_E is fixed; m_0 , λ , the D1/D4 scale τ , and on the accidental child the D2 amplitude z range over their accepted compact or interval objects. The protocol coordinate contains the finite physical geometry, record alphabet, control design, diagonal and squared-modulus reconstruction relations, finite exposures, confidence level, and frozen falsifier of P7-S1 through P7-S2.

Theorem 8.2 (P7-12 – parameter-status theorem). *Every parameter entering a PCR-7 observable has exactly one status: fixed structural, calibrated anchor, standing variable, bounded source variable, nuisance, out of scope, or target update. In particular,*

$$s_E \in \Theta_{\text{fixed}}, \quad \lambda \in I_E, \quad (m_0, \tau, z) \in \Theta_{\text{standing/bounded}},$$

and no lawful ready-child path sends s_E to zero.

Proof. The PCR-5E source-qualified comparison certificate fixes s_E after the positive product-box theorem and before joint cardinality. Varying it changes the E-H2 child source point and therefore is not an internal PCR-7 nuisance. The threshold λ , light scale m_0 , common D1/D4 scale τ , and accidental D2 amplitude z are explicit lawful coordinates. Detector quantities are protocol nuisances formed independently in P7-S1. Exposure and confidence level are frozen design coordinates established by P7-S2A/P7-S2, not fitted source parameters. Any unlisted source coefficient or new response is an update. The categories are disjoint by provenance and target use. $\square$

Theorem 8.3 (P7-13 – physical detector and nuisance completion). *Let*

$$p_{\star} := p_{\star}^{\text{fin}}$$

be the finite source-authorized protocol of P7-S1–P7-S2. Its independently fixed nuisance category includes parent-flux calibration, flavor tagging, fiducial and decay-volume acceptance, mass response, backgrounds, diagonal reconstruction, squared-modulus null certification, pole reconstruction, and finite statistical uncertainty. The complete certificate

$$\text{Cert}_{p_{\star}} = (H(I_E), J_Q, \mathcal{K}_{\star}, a_{\star}, \Xi_{\text{strat}}, L_{\text{diag}}, L_{\text{quad}}, E_{\star}, \text{Cal}, \text{Bkg}, \text{Rec}, \mathbf{n}_{\text{diag}}, \mathbf{n}_{\text{quad}}, \alpha_{\star}, \eta_{\text{diag}}, \eta_{\text{quad}}, \varepsilon_{\star})$$

has full heavy-window coverage, a derived acceptance lower bound $a_{\star} > 0$, finite diagonal and quadratic response constants, finite predeclared sample sizes, one-sided null bounds

$$q^U < \eta_{\text{quad}} = \left(\frac{\delta_E}{16} \right)^2,$$

and confidence bounds

$$0 \leq \varepsilon_\star \leq \frac{3}{8} \delta_E < \frac{\sqrt{6}}{2} \delta_E.$$

Zero acceptance, loss of source/process coverage, loss of stratified identifiability, an unmodelled source-coherent pole, or a change of response principle is not a lawful nuisance point.

Proof. P7-S1 constructs the physical source, finite detector geometry, positive-response cover, common carrier/reference record, and finite calibrated nuisance model. P7-S1A supplies finite diagonal and squared-modulus reconstruction constants without a false amplitude-level inverse. P7-S1B gives the rank-three certificate from diagonal lower bounds and one-sided quadratic null bounds. P7-S2A supplies finite signal and control sample sizes and derives both the strict diagonal confidence margin and, through P7-S1B, the complete matrix-packet radius; P7-S2 freezes the complete protocol. Thus neither positive acceptance nor the precision bounds are inserted as protocol axioms. The nuisance coordinates are exactly the finite calibrated fields entering the common response model. A zero-efficiency or stratified-nonidentifiable point fails the physical protocol certificate; an unmodelled process changes the response/source class and is routed to NotReady or update. Hence the nuisance object is complete at its stated scope. $\square$

Definition 8.4 (Decoupling and hiding). A candidate observable is decouplable when

$$\text{cl}(\mathcal{R}_{b,p}^{\text{car}}) \cap \mathcal{R}_{b,p}^{\text{ref}} \neq \emptyset$$

under a target-preserving lawful path. Limits that leave the ready child, remove a mandatory source incidence, or invalidate the physically realized stratified protocol are updates/inapplicability, not decoupling routes.

Theorem 8.5 (P7-14 – decoupling topology theorem). *The lawful decoupling atlas consists of source-coordinate limits inside the ready child, threshold motion inside I_E , enriched D1/D4 and D2 limits, lawful groupoid arrows, and detector/nuisance limits inside the certified physical protocol. It contains no $s_E \rightarrow 0$ path, no H_λ path outside the scanned compact interval, and no zero-acceptance or stratified-nonidentifiable detector point.*

Proof. PCR-5E fixes the child parameter domains and their closures. The source variables are those of P7-12. The detector object is the physically realized compact nuisance/record family of P7-S1–P7-S2/P7-13. Source topology and positive vectorlike incidence remain fixed. A path with $s_E \rightarrow 0$ crosses the source-activity/child comparison boundary; a path with unbounded H_λ leaves I_E ; a path with vanishing physical acceptance or loss of the stratified record relation leaves $p_\star$. The remaining paths are exactly the listed lawful continuations. $\square$

8.1 Compact parameter domains and response envelopes

Lemma 8.6 (Compact response envelope). *For the primary packet, the lawful source coordinates affecting the ideal response form a compact object: $\lambda \in I_E$, while δ_E is fixed and all remaining child coordinates factor out. With the compact physical nuisance/control object of P7-S1, the carrier and reference stratified record families are compact and disjoint, their diagonal-record distance is positive, and the discrete pole/rank separator is attained.*

Proof. The heavy interval is compact by the E-S4A product-box theorem. The packet is independent of m_0, τ, z and lawful unitary presentation. P7-S1 forms finitely many compact kinematic, geometry, detector, calibration, and background coordinates, and P7-S2 freezes finite exposures and confidence rules. The expected stratified-record map is continuous on each applicability cell and the protocol remains in one complete cell. Continuous images of compact objects are compact. P7-S1A makes the stratified carrier/reference quotient injective, and P7-S2A gives a positive diagonal confidence margin; the pole classes are separately disjoint. The separation is therefore positive and attained. $\square$

8.2 Nuisance transformations and quotient structure

Let G_p^{nuis} denote the lawful nuisance groupoid generated by calibration, efficiency, resolution, binning, reconstruction, background, and finite-sample coordinates. The raw response space is acted on before quotienting:

$$G_p^{\text{nuis}} \curvearrowright \text{Rec}_p, \quad \text{QRec}_p = \text{Rec}_p / \sim_{\text{det}}.$$

Proposition 8.7 (Nuisance soundness). *A nuisance transformation may move a raw count vector while preserving the stratified reconstructed packet. It may not alter source-coherent pole support, the certified rank-three class, the diagonal carrier/reference separation, or the ideal equality class $\Delta_{\text{act}} = \Omega_H$ without leaving the declared response model or the predeclared confidence relation.*

Proof. The quotient is defined by complete detector-use agreement. P7-S1 supplies a source-to-record map and common controls; P7-S1A/P7-S1B define the target-adapted record quotient and rank certificate; P7-S2A/P7-S2 supply the finite simultaneous confidence relation. Lawful calibration and finite-sample fluctuations therefore transport to the same stratified packet class. Changing a retained pole/rank/diagonal value beyond the frozen relation changes an admitted detector use. If caused by an unmodelled response, it is NotReady or update rather than quotient-trivial nuisance motion. $\square$

9 Cross-child charged-current closure invariant

9.1 Source-level identities

For every ready child and accepted realization, the intrinsic charged-current blocks are

$$N_X = c_E U_X, \quad V_{H,X} = -s_E U_X, \quad \delta_E = s_E^2,$$

with U_X unitary. The heavy mass block is $H_\lambda \mathbf{1}_3$.

Definition 9.1 (Closure packet). Define

$$\begin{aligned} \Delta_{\text{act}}(X) &:= \mathbf{1}_3 - N_X N_X^\dagger, & \Omega_H(X) &:= V_{H,X} V_{H,X}^\dagger, \\ C_H(X) &:= \Delta_{\text{act}}(X) - \Omega_H(X), \end{aligned}$$

and let $\text{Pole}_H(X)$ record the detector-quotient heavy support, multiplicity/rank class, and mass-window location. Let $\mathcal{A}_{\mu e}^H(X)$ be the heavy contribution to the $\mu \rightarrow e\gamma$ response.

Theorem 9.2 (P7-15 – common-core/decoration observable decomposition). *The packet*

$$O_\star(X) = (\Delta_{\text{act}}(X), \Omega_H(X), C_H(X), \text{Pole}_H(X), \mathcal{A}_{\mu e}^H(X))$$

factors entirely through the PCR-6 common carrier core. It is independent of the accidental/exact-global decoration. LNV amplitude and protector records form a separate decoration packet.

Proof. The blocks N_X , $V_{H,X}$, the heavy mass map, and the unitary matrix are common to both children. The accidental/exact-global difference lies in the D2 amplitude and D5 protector decoration, neither of which enters the charged-current Gram matrices or heavy pole. The heavy CLFV cancellation follows from the same common block. Hence the displayed packet factors through the common core, while the LNV/protector packet remains branch-indexed. $\square$

Theorem 9.3 (P7-16 – invariant-observable theorem). *For all $b \in \mathcal{B}_7^{\text{ready}}$ and $X \in \text{PhysMod}_{E,\text{VJ}}^b$,*

$$\Delta_{\text{act}}(X) = \Omega_H(X) = \delta_E \mathbf{1}_3, \quad C_H(X) = 0.$$

Moreover,

$$\text{rank } \Omega_H(X) = 3, \quad \text{Pole}_H(X) = \text{Deg}_3(H_\lambda), \quad H_\lambda \in H(I_E),$$

and

$$\mathcal{A}_{\mu e}^H(X) = 0.$$

These identities are natural across both ready child groupoids and constant under all lawful nonprotocol parameters except for the pole location inside the predicted compact class.

Proof. Unitarity gives

$$N_X N_X^\dagger = c_E^2 U_X U_X^\dagger = c_E^2 \mathbf{1}_3,$$

so $\Delta_{\text{act}} = (1 - c_E^2) \mathbf{1}_3 = s_E^2 \mathbf{1}_3$. Likewise

$$V_{H,X} V_{H,X}^\dagger = s_E^2 U_X U_X^\dagger = \delta_E \mathbf{1}_3.$$

The exact vectorlike mass map has one rank-three degenerate heavy block, giving the pole/rank class. Finally,

$$\mathcal{A}_{\mu e}^H \propto F(H_\lambda^2/m_W^2) \sum_{k=1}^3 (V_H)_{ek} (V_H)_{\mu k}^* = \delta_E F \sum_k U_{ek} U_{\mu k}^* = 0.$$

The variables m_0, τ, z do not enter these expressions; λ moves only the pole inside $H(I_E)$; lawful $U(3)$ changes cancel in the Gram matrices. Branch decoration is absent by P7-15. Thus the theorem is uniform. $\square$

Theorem 9.4 (P7-17 – exact selection-rule theorem). *The record predicates*

$$\text{RankHeavy} = 3, \quad \text{ClosureMatched} : \Delta_{\text{act}} = \Omega_H, \quad \text{HeavyGIM}_{\mu e} = 0$$

are exact target-relative selection rules on both ready children.

Proof. They follow from P7-16 and are invariant under lawful detector quotienting because rank, equality of calibrated Gram matrices, and the zero of the complete heavy interference sum are target-admitted record uses. No numerical mass, basis, or source label is required. $\square$

Theorem 9.5 (P7-R2 – minimal observable packet theorem). *The minimal cross-child prediction packet is*

$$O_{\star}^{\min} = (\Delta_{\text{act}}, \Omega_H, \text{Pole}_H).$$

Deleting any one of its three components destroys either reference separation, source matching, or detector completeness. The closure residual and heavy CLFV zero are derived checks rather than minimal separating coordinates.

Proof. Without Δ_{act} , a heavy-like background is not connected to the active closure defect. Without Ω_H , a normalization deficit can be absorbed into the reference detector/calibration family. Without Pole_H , an abstract heavy-weight matrix lacks a detector support class and can be represented as an incomplete response. Together the three identify a nonzero source-matched heavy support. The equality residual is determined by the first two, and the CLFV zero does not separate the no-carrier reference, so neither is minimal. $\square$

9.2 Basis-free formulation

The packet can be written without choosing the unitary matrix U . Let ι_L and ι_H be the light and heavy charged-current incidence maps from the boundary flavor object to the propagation carrier. Then

$$N = \Pi_L \iota_L, \quad V_H = \Pi_H \iota_H,$$

and the two positive forms are

$$G_L = N N^\dagger, \quad G_H = V_H V_H^\dagger.$$

The source mass-map factorization implies

$$G_L + G_H = \mathbf{1}_3, \quad G_H = \delta_E \mathbf{1}_3.$$

This is the basis-free active/heavy closure identity.

Lemma 9.6 (Lawful groupoid invariance). *Under every lawful source/boundary arrow (W_L, W_H, W_∂) , the forms transform by common boundary conjugation,*

$$G_{L,H} \mapsto W_\partial G_{L,H} W_\partial^\dagger,$$

and therefore their spectrum, rank, equality relation, and scalar class $\delta_E \mathbf{1}_3$ are invariant.

Proof. The incidence maps transform covariantly and the internal unitary factors cancel between a map and its adjoint. The boundary quotient identifies common conjugation. Since the forms are scalar matrices at the source point, they are fixed pointwise. $\square$

9.3 Heavy-pole multiplicity as an observable class

Lemma 9.7 (Heavy spectral-support descent). *The threefold heavy eigenspace descends under the lawful $U(3)_H$ action to one pole-support object with multiplicity/rank three. The protocol record does not depend on a choice of heavy eigenbasis.*

Proof. PCR-6 proves complete $U(3)_H$ witness descent. The detector protocol uses the inclusive spectral measure and boundary Gram form, both invariant under heavy-subspace rotation. It therefore records one pole location with rank-three weight, not three labeled particles. $\square$

9.4 Source-matching identity

Proposition 9.8 (Closure sum rule). *The exact relation*

$$\Delta_{\text{act}} = \Omega_H$$

is equivalent, at the ready child source, to conservation of the complete charged-current boundary norm between the light and heavy subspaces.

Proof. The full light-heavy block rotation is unitary. Hence the sum of the two boundary Gram forms is identity. Rearranging gives the relation. Conversely, positivity together with the fixed light/heavy source ranks reconstructs the same norm decomposition at the observable level. $\square$

10 No-vectorlike reference responses

Theorem 10.1 (P7-11 – reference response completeness). *For every reference $R_0 \in \{R_0^{\text{U3}}, R_0^{\text{SM0}}\}$ and every lawful reference nuisance/continuation,*

$$\Delta_{\text{act}}(R_0) = 0, \quad \Omega_H(R_0) = 0, \quad \text{rank } \Omega_H(R_0) = 0, \quad \text{Pole}_H(R_0) = \text{NoHeavyPole}.$$

The reference envelope contains no record with a coherent rank-three heavy pole matched to a nonzero active deficit.

Proof. By construction $V_{H,0} = 0$. The unitary reference has $N_0 N_0^\dagger = \mathbf{1}_3$; the minimal boundary reference has no vectorlike closure defect in the declared protocol. Detector and background continuations may alter raw counts but cannot introduce a source-coherent heavy pole or a calibrated nonzero closure matrix without adding a new source or response class. Such an addition is an explicit reference update, not a nuisance point. The stated envelope is therefore complete. $\square$

Theorem 10.2 (P7-R3 – reference robustness theorem). *The common packet separation is invariant under every reference presentation preserving the known substrate, unitary/no-vectorlike source status, detector, background class, calibration map, and protocol. Any reference capable of reproducing $\mathcal{R}_\star$ contains a new heavy source or nonunitary carrier and is therefore outside the no-vectorlike reference class.*

Proof. All lawful reference presentations descend to the same complete record trace of P7-11. A record in $\mathcal{R}_\star$ has nonzero Ω_H , rank three, and a heavy pole. These cannot arise by gauge, basis, chart, binning, or calibration redescription of a zero heavy source. Therefore a mimicking reference changes source content and is a target update. $\square$

Remark 10.3. Reference robustness is not uniqueness of microscopic explanation. A richer theory containing another carrier could yield the same packet; it would be a different non-reference target and would require a separate comparison.

10.1 Lawful reference presentations

The reference atlas permits lawful changes of light mixing basis, calibration chart, background parametrization, and detector reconstruction. It does not permit a nonzero heavy spectral measure or a nonunitary active source.

Lemma 10.4 (Reference closure under detector transport). *If a raw reference record contains a fluctuation or background feature inside $H(I_E)$, it remains in the reference envelope only when the complete correlation packet fails the source-matching conditions*

$$\Omega_H = \Delta_{\text{act}}, \quad \text{rank } \Omega_H = 3, \quad \text{FlavorUniversal} = 1.$$

Proof. Reference backgrounds are carried through the same joint detector quotient. A local excess can occur in one channel, but the reference source has $\Delta_{\text{act}} = 0$ and no common heavy incidence. Producing the full packet requires adding source-coherent correlated responses across the active/heavy records, which is a new source rather than a background presentation. $\square$

Proposition 10.5 (No post-hoc reference broadening). *After $\mathcal{R}_\star$ is fixed, the reference family cannot be enlarged by adding arbitrary nonunitary or heavy-pole models while retaining the name “no-carrier reference.”*

Proof. The reference fingerprint was frozen at P7-10. Adding a response specifically to overlap the prediction changes source content and comparison target. Such an alternative may be studied as a competing carrier theory but not as a same-reference nuisance. $\square$

11 Detector-level confidence separation

Construction 11.1 (Flavor-complete charged-current spectral tomography). The primary protocol is the physical tagged- W^* dual-arm charged-current protocol $p_\star^{\text{fin}}$ of P7-S1. Its reconstructed packet has four stages:

1. source-normalized active charged-current tomography in all three boundary flavors, using tagged charged-current parent rates and the finite short-baseline interaction array;
2. flavor-complete heavy production/analyzer records in the stratified coordinates (d^H, q^H) and an invariant-mass scan of the compact heavy window $H(I_E)$;
3. common control calibration and reconstruction of the diagonal coordinates together with one-sided upper bounds on the predicted-null squared-modulus coordinates;
4. a joint quotient record retaining the heavy pole/rank class and the ideal equality residual.

The future validation packet uses the full correlated record. Upstream electron-active information appears only in the calibration/derivation block, not as held-out evidence.

Definition 11.2 (Stratified carrier confidence class). Let

$$\eta_d = \frac{\delta_E}{8}, \quad \eta_q = \left(\frac{\delta_E}{16}\right)^2.$$

The carrier confidence relation $\widehat{\mathcal{R}}_{\star, \eta_d, \eta_q}^{\text{strat}}$ consists of all positive-semidefinite pairs (Δ, Ω) and pole classes satisfying

$$\begin{aligned} |\Delta_{\alpha\alpha} - \delta_E| &< \eta_d, & |\Omega_{\alpha\alpha} - \delta_E| &< \eta_d, \\ |\Delta_{\alpha\beta}|^2 &< \eta_q, & |\Omega_{\alpha\beta}|^2 &< \eta_q \quad (\alpha < \beta), \end{aligned}$$

with one rank-three degenerate pole in $H(I_E)$. The reference confidence relation $\widehat{\mathcal{R}}_0^{\text{strat}}$ is defined by the same record quotient with all six diagonal coordinates within η_d of zero, the same one-sided quadratic bounds, rank zero, and no heavy pole.

Theorem 11.3 (P7-18 – ratio, shape, and bounded-region theorem). *Under $p_\star$, every carrier record lies, with simultaneous coverage at least $1 - \alpha_\star$, in $\widehat{\mathcal{R}}_{\star, \eta_d, \eta_q}^{\text{strat}}$. The ideal ratio and shape relations are*

$$\frac{(\Delta_{\text{act}})_{\alpha\alpha}}{(\Delta_{\text{act}})_{\beta\beta}} = 1, \quad \frac{(\Omega_H)_{\alpha\alpha}}{(\Omega_H)_{\beta\beta}} = 1, \quad \Delta_{\text{act}} = \Omega_H,$$

for all flavors, with zero off-diagonal squared-modulus coordinates in the boundary-covariant quotient. Every matrix pair in the carrier confidence relation has rank three by P7-S1B.

Proof. P7-16 fixes the ideal matrices. P7-S1 constructs the finite physical record; P7-S1A identifies the diagonal and squared-modulus quotient coordinates; P7-S1B converts the diagonal and one-sided quadratic bounds into the rank-three certificate; and P7-S2A/P7-S2 provide the frozen simultaneous confidence relation. Ratios and ideal equality are invariant under common normalization and lawful flavor-basis transport. The pole/rank class is retained by the record quotient. Thus every lawful carrier record lies in the displayed class at the predeclared coverage. $\square$

Definition 11.4 (Diagonal separation projection). Define

$$\pi_d(\Delta, \Omega) := ((\Delta_{\alpha\alpha})_{\alpha=e,\mu,\tau}, (\Omega_{\alpha\alpha})_{\alpha=e,\mu,\tau}) \in \mathbb{R}^6.$$

On the carrier class,

$$\pi_d = (\delta_E \mathbf{1}_3, \delta_E \mathbf{1}_3),$$

whereas on the no-vectorlike reference class, $\pi_d = 0$.

Theorem 11.5 (P7-19 – branch-indexed physical-record separation). *For each $b \in \mathcal{B}_7^{\text{ready}}$, the carrier and reference stratified confidence relations under the physically realized protocol $p_\star$ are disjoint. On the ideal continuous matrix coordinates the Frobenius gap remains*

$$\text{Gap}_{b,p_\star}^{\text{ideal}} = \sqrt{6} \delta_E.$$

On the reconstructed diagonal projection, P7-S2A gives

$$\text{Gap}_{b,p_\star}^{\text{diag}} \geq \sqrt{6} \delta_E - 2\varepsilon_\star^{\text{diag}} \geq \frac{3\sqrt{6}}{4} \delta_E > 0.$$

The complete continuous matrix confidence classes additionally satisfy

$$\text{Gap}_{b,p_\star}^{\text{mat}} \geq \sqrt{6} \delta_E - 2\varepsilon_\star \geq \left(\sqrt{6} - \frac{3}{4}\right) \delta_E > 0$$

with simultaneous coverage at least $1 - \alpha_\star$. P7-S1B and the pole-support record supply independent rank and pole disjointness certificates.

Proof. The Frobenius norm of $\delta_E \mathbf{1}_3$ is $\sqrt{3}\delta_E$. Since both Δ_{act} and Ω_H differ from zero, the ideal product-space distance is

$$\sqrt{3\delta_E^2 + 3\delta_E^2} = \sqrt{6}\delta_E.$$

The same value is the Euclidean separation of the six-dimensional diagonal vectors. P7-S2A bounds each reconstructed diagonal class by radius at most $\varepsilon_\star^{\text{diag}} \leq \sqrt{6}\delta_E/8$, so the triangle inequality gives the first positive gap. P7-S1B places each complete matrix pair within product-space radius $\varepsilon_\star \leq 3\delta_E/8$ of its ideal center, which gives the second positive gap. It also shows that every carrier matrix pair in the confidence relation has rank three, while the reference has rank zero/no pole. Therefore the full stratified confidence relations do not overlap. $\square$

Theorem 11.6 (P7-20 – cross-child common-core separation). *The same physical protocol $p_\star$, source/control geometry, stratified record class, reference class, nonlinear finite-exposure certificate, and separator work simultaneously for b_{acc} and b_{glob} .*

Proof. By P7-15 the packet factors through the common carrier core, and by P7-16 its value is independent of branch decoration. P7-11 gives the same reference class. P7-S1 uses only the common charged-current source, active/heavy propagation, and detector controls; it contains no LNV/protector-specific operation. P7-S1A/P7-S1B use the same stratified quotient and rank criterion, and P7-S2A/P7-S2 depend only on the common compact response and the fixed δ_E . Hence the branchwise separation theorems have identical protocol data and combine into one cross-child result. $\square$

11.1 Detector transport as a statistical relation

Let $\mathcal{D}_{p_\star}^{(\mathbf{n}_d, \mathbf{n}_q)}$ be the finite-exposure detector relation from ideal packet space to the stratified confidence quotient. The protocol certificate requires response soundness, one-sided null coverage, and simultaneous coverage:

$$r \in \mathcal{D}_{p_\star}^{(\mathbf{n}_d, \mathbf{n}_q)}(q) \implies q \in \mathcal{C}_{p_\star}^{\text{strat}}(r), \quad \Pr_q[q \in \mathcal{C}_{p_\star}^{\text{strat}}(R)] \geq 1 - \alpha_\star.$$

Lemma 11.7 (Transported diagonal-gap inequality). *If ideal packet classes A, B have diagonal-projection distance g_d and both reconstructed diagonal confidence classes have radius at most $\varepsilon_d < g_d/2$, then the complete stratified confidence relations are disjoint whenever their rank/pole labels are also disjoint or equal only within the same target class. Their diagonal distance is at least $g_d - 2\varepsilon_d$.*

Proof. For $a \in A, b \in B$ and reconstructed diagonal representatives a', b' , the triangle inequality gives

$$\|\pi_d(a') - \pi_d(b')\| \geq \|\pi_d(a) - \pi_d(b)\| - \|\pi_d(a) - \pi_d(a')\| - \|\pi_d(b) - \pi_d(b')\| \geq g_d - 2\varepsilon_d > 0.$$

The one-sided quadratic bounds and pole/rank labels are retained as additional coordinates, so nonoverlap of the diagonal projection already implies nonoverlap of the full stratified relation. P7-S2A supplies the finite-exposure bounds used in P7-19. $\square$

Proposition 11.8 (No detector-null direction in the stratified quotient). *On the carrier/reference comparison family, no nonzero variation of the retained diagonal, squared-modulus, or pole-support coordinates lies in the kernel of the complete physical response and control system. Phase directions not represented by these coordinates are intentionally quotient-trivial and are not claimed to be measured.*

Proof. This is the identifiability conclusion of P7-S1A. The light-sampling transform, heavy diagonal production blocks, heavy squared-modulus analyzer blocks, pole/background blocks, and common calibration controls have positive finite response coefficients on their retained quotient coordinates. A record-null variation therefore vanishes in every component of Ξ_{strat} . No conclusion is drawn about phase directions outside the declared quotient. $\square$

12 Uniformity and nondecoupling across completed sectors

Theorem 12.1 (P7-21 – nondecoupling theorem). *No lawful path in the ready child closure sends the $p_\star$ carrier envelope to the reference envelope. Equivalently,*

$$\text{cl}(\mathcal{R}_{b,p_\star}^{\text{car}}) \cap \mathcal{R}_{b,p_\star}^{\text{ref}} = \emptyset \quad \forall b \in \mathcal{B}_7^{\text{ready}}.$$

Proof. The packet depends only on $\delta_E = s_E^2$, the compact heavy interval, and the common source incidence. The parameters m_0, τ , and z do not enter. Variation of λ moves a rank-three pole within a completely scanned compact interval and does not change its inclusive weight. Unitary/source presentation changes cancel in the Gram matrices. The only algebraic route to $\Delta_{\text{act}} = \Omega_H = 0$ is $s_E \rightarrow 0$, but P7-12 shows that s_E is fixed and that the zero limit leaves the source-qualified child. The only detector routes to zero are loss of mass-window coverage, acceptance, or response completeness. P7-S1 proves positive physical acceptance on the complete interval, P7-S1A proves stratified record identifiability, P7-S1B certifies rank from one-sided quadratic null bounds, P7-S2A/P7-S2 prove finite nonlinear confidence and a certified margin, and P7-13 therefore classifies their loss as protocol failure/inapplicability rather than a lawful nuisance limit. Therefore no lawful target-preserving path reaches the reference. $\square$

Theorem 12.2 (P7-R4 – cross-child prediction stability theorem). *The prediction-lock packet and its falsifier are natural under every lawful arrow within either child and are identical under the common-core comparison between the accidental and exact-global children.*

Proof. Naturality is P7-08. The common-core comparison preserves N, V_H, s_E , and the heavy pole class while changing only the D2/D5 decoration. The packet and reference are therefore transported identically. The complement of the common predicted class in the same detector quotient is the same falsifier set. $\square$

Corollary 12.3 (Uniform positive-margin lock). *For the physically realized protocol $p_\star$ of P7-S1–P7-S2/P7-13,*

$$\inf_{b \in \mathcal{B}_7^{\text{ready}}} \text{dist}(\mathcal{D}_{p_\star}(\mathcal{R}_{b,p_\star}^{\text{car}}), \mathcal{D}_{p_\star}(\mathcal{R}_{b,p_\star}^{\text{ref}})) \geq \sqrt{6}\delta_E - \frac{3\delta_E}{4} > 0.$$

12.1 Route-by-route hiding audit

Proposed hiding route	Would-be limit	Exact disposition
Mixing decoupling	$s_E \rightarrow 0$	Leaves fixed E-H2 child source/activity certificate.
Threshold decoupling	$H_\lambda \rightarrow \infty$	Leaves compact I_E ; $p_\star$ covers the complete interval.
Flavor rotation	Change U	Gram matrices unchanged.
Absolute light scale	Vary m_0, τ	Packet independent.
LNV decoration	Vary z	Packet independent.
Global/accidental decoration	Change protector status	Common packet independent.
Acceptance hiding	Efficiency $\rightarrow 0$	Contradicts the derived positive detector-cell response of P7-S1 or leaves the protocol.
Calibration shift	Absorb Δ_{act}	Fails matched heavy-weight and pole relations.
Background mimicry	Insert a peak	Fails full flavor/closure packet or changes reference source.
Resolution smearing	Merge pole with continuum	Excluded by the finite calibrated pole partition of P7-S2; loss of that partition leaves the protocol.

Corollary 12.4 (No target-preserving hiding path). *Every route capable of meeting the reference either leaves the ready child, leaves the complete protocol, or changes the reference source. Hence none belongs to the lawful decoupling closure used by the prediction theorem.*

13 Target-dependent observables and residual channels

The common packet does not exhaust all experimentally relevant channels. The remaining channels are classified separately so that upstream inputs and decouplable marginal responses do not inherit the authority of the common lock.

13.1 LNV and protector decoration

Proposition 13.1 (One-sided branch discriminator). *On the exact-global child the physical D2 amplitude is the defined zero subfibre. On the accidental child the amplitude is $zA_{76}^\star$ with z in its accepted interval. Therefore a future independently formed, target-faithful nonzero LNV record excludes the exact-global child, while a null result does not select either child unless the accidental interval is separately proved bounded away from zero.*

Proof. This is the PCR-5E obstruction relation transported to a new response protocol. Exact-global protection forbids active LNV response at the child scope. The accidental target permits it but does not, from the available source interval alone, force a nonzero held-out response. Thus the implication is one-sided. $\square$

13.2 Residual observable channels

Route	Complete disposition	Prediction status
D1 beta endpoint	Target/calibration input; varies with m_0, τ	Not held out; derived cross-channel use only.
D2 same-isotope LNV	Target input and branch decoration	Not held out; another protocol can discriminate one-sidedly.
D3 existing direct-state charts	Used to certify the child and protocol region	Cannot validate themselves.

Route	Complete disposition	Prediction status
D3 active tester C_{ee}	Derivation input fixing child acceptance	Electron value is not held out; flavor-complete closure relation is new.
Heavy CLFV $\mu \rightarrow e\gamma$	Exact heavy zero; light GIM term inherited	Overlaps no-carrier reference; consistency check, not separation.
D4 public mass sum	Target input; common D1/D4 generator	Not a prediction.
B-L protector/action	Structural distinction	Requires an independently formed experimental protocol; exact-global nonzero LNV is excluded.
Mass ordering	Trace/exclusion realization absent	No pre-lock prediction at current certificate.
Individual heavy rate	Protocol and threshold dependent	Conditional unless embedded in $p_\star$ inclusive packet.

Theorem 13.2 (P7-24 – no-lock route exhaustion outside the common packet). *Within the declared observable atlas, every route not contained in $O_\star^{\min}$ is exactly one of: upstream input, derived consistency relation, branch-only discriminator, conditional response, decouplable/overlapping channel, or NotReady. No second cross-child prediction lock is established.*

Proof. P7-09 exhausts the atlas. The preceding table follows from the D1–D5 and ordering theorems. The only cross-child source relation with an independently formed held-out protocol, complete detector response, reference separation, and nondecoupling is the active/heavy closure packet. All other routes fail at least one lock premise. Hence no second lock follows. $\square$

13.3 Why no single rate carries the full theorem

A direct heavy production rate depends on beam energy, parton or nuclear response, lifetime, geometry, acceptance, and the pole location. Those quantities can make a channel experimentally weak without altering the source’s inclusive charged-current weight. The packet theorem therefore reconstructs the source-normalized Gram form and pole support rather than asserting one universal raw event count. This distinction is the observable analogue of the earlier carrier/interface separation.

Proposition 13.3 (Individual-rate nonuniversality). *No single unnormalized direct-state event rate is constant over the full child/protocol atlas. Any such rate is at most a branch-stable bounded or conditional response unless normalized into the complete $p_\star$ packet.*

Proof. The physical rate depends continuously on H_λ and detector/beam response. Since λ varies, nominal rates vary. The inclusive source weight remains fixed but is not equal to a raw count until calibrated response transport. Therefore a universal raw-rate claim would ignore lawful protocol dependence. $\square$

Proposition 13.4 (CLFV-null nonseparation). *The exact heavy contribution $\mathcal{A}_{\mu e}^H = 0$ is a robust consistency prediction but not a reference separator, because the no-vectorlike references also have zero heavy contribution.*

Proof. Both carrier and reference record classes contain the same zero value in that marginal coordinate. Hence the closures overlap. The coordinate strengthens source matching inside the joint packet but cannot alone satisfy P7-19. $\square$

14 Experimental prediction lock

Definition 14.1 (Prediction lock). For branch b and protocol p ,

$$\text{PredLock}_{b,p} \iff \text{Invariant}_{b,p} \wedge \text{ReferenceSeparated}_{b,p} \wedge \text{Nondecoupling}_{b,p} \\ \wedge \text{ProtocolComplete}_p \wedge \text{DataFirewall}_p.$$

Here $\text{ProtocolComplete}_p$ requires an actually constructed source-to-record response, uniform positive acceptance, stratified diagonal/quadratic/pole identifiability, common carrier/reference calibration, a finite predeclared exposure, and a simultaneous confidence/falsifier rule. A cross-child lock requires the same packet, physical protocol, predicted class, separator, and confidence rule for every $b \in \mathcal{B}_7^{\text{ready}}$.

Theorem 14.2 (P7-22 – prediction-strength classification). *The observable atlas receives the deterministic statuses*

$$\begin{aligned} O_{\star}^{\min} &: \text{CrossChildJointPacketPredictionLock}, \\ \text{independent nonzero LNV record} &: \text{OneSidedBranchFalsifier}, \\ \text{heavy CLFV zero} &: \text{StructuralOnly/Overlapping}, \\ \text{mass ordering} &: \text{NoPreLockPrediction}, \\ \text{target-used D1–D4 records} &: \text{InputNotHeldOut}, \\ \text{uncompleted protocols} &: \text{NotReady}_{\text{protocol}}. \end{aligned}$$

Proof. Apply the five lock predicates to every atlas entry. P7-S1A/P7-S1B/P7-S2A and their summary theorems P7-S1/P7-S2 establish stratified physical-protocol realization, one-sided quadratic null certification, rank closure, and finite-exposure completeness for the common packet; P7-16–P7-21 establish invariance, exact packet form, reference separation, and nondecoupling. The remaining entries fail exactly as stated in P7-24. The classifier is deterministic and target-tagged. $\square$

Theorem 14.3 (P7-23 – cross-child finite-protocol prediction-lock theorem). *The physically realized protocol $p_{\star} = p_{\star}^{\text{fin}}$ and packet $O_{\star}^{\min}$ satisfy*

$$\text{PredLock}_{p_{\star}}^{\text{cross}}.$$

At the predeclared simultaneous coverage $1 - \alpha_{\star}$, the carrier and no-vectorlike reference confidence regions are disjoint. Therefore

$$\text{Status}(\text{PCR-7}; \mathcal{B}_7^{\text{ready}}) = \text{CrossChildJointPacketPredictionLock}_{\text{CCST}}.$$

Proof. P7-16 gives exact invariance and P7-08 gives moduli naturality. P7-S1 constructs a finite source-authorized physical response with uniform positive acceptance, common controls, and identifiable diagonal/quadratic/pole records. P7-S2 derives finite exposures with the correct nonlinear null scaling, a frozen diagonal radius $\varepsilon_{\star}^{\text{diag}} \leq \sqrt{6}\delta_E/8$, and a complete set-valued matrix-packet radius $\varepsilon_{\star} \leq 3\delta_E/8 < \sqrt{6}\delta_E/2$. P7-19/P7-20 therefore give branchwise and cross-child physical-record separation. P7-21 gives nondecoupling, and P7-02 gives the held-out firewall. These conditions are necessary by definition and jointly sufficient because the complete detector quotient and confidence rule are now constructed rather than postulated. Hence the prediction lock follows. $\square$

Theorem 14.4 (P7-R5 – prediction authority and claim-language theorem). *The strongest licensed prose is: “At the two completed vectorlike-joint child scopes, a finite source-authorized tagged charged-current protocol predicts the common active/heavy closure packet and a rank-three heavy pole; for the predeclared finite exposure, the carrier and no-vectorlike reference confidence regions are disjoint at coverage $1 - \alpha_{\star}$.” The theorem does not license “the packet has been observed,” “the vectorlike source is uniquely identified,” “an existing facility has already implemented the complete protocol,” or “all neutrinos have this observable structure.”*

Proof. The first sentence restates P7-S1A/P7-S1B/P7-S2A/P7-S1/P7-S2/P7-23 with their target, physical-protocol, exposure, and confidence tags. Empirical confirmation is absent by P7-02/P7-27; uniqueness of microscopic explanation is denied by P7-R3; present implementation is excluded by the physical-realizability remark following P7-S2; and universalization is blocked by P7-00/P7-01. Stronger language exceeds at least one theorem premise. $\square$

14.1 Necessity and sufficiency

Proposition 14.5 (Necessity). *Each of invariance, reference separation, nondecoupling, physical protocol completeness, and data-firewall closure is necessary for an unavoidable experimental prediction at fixed child scope.*

Proof. Without invariance, lawful realizations disagree. Without reference separation, the record does not distinguish carrier and reference. Without nondecoupling, a lawful realization hides the response. Without physical protocol completeness, the detector-level class is undefined, unidentifiable, or unattainable at finite exposure.

Without the data firewall, the claim is calibrated or selected by its alleged validation. Each failure therefore defeats prediction authority. $\square$

Proposition 14.6 (Sufficiency). *At a complete physical detector-record quotient, the conjunction of the five criteria is sufficient for a target-qualified prediction lock.*

Proof. Invariance places every lawful carrier response in one predicted class. Reference separation excludes every lawful reference response. Nondecoupling closes source, parameter, and nuisance boundaries. P7-S1A/P7-S1B/P7-S2A and P7-S1/P7-S2 make the protocol experimentally well typed and finitely testable with a frozen stratified confidence rule. The firewall makes the comparison pre-lock. Thus every lawful future complete record either lies in the predicted confidence class or enters the predeclared falsifier/update classification. $\square$

14.2 Strength of the result

The original ARC asks for an observable common to all lawful carrier realizations and separated from the no-carrier reference after parameter, decoupling, and detector analysis. P7-23 proves that baseline and strengthens it: the observable is a finite higher-arity packet, common to two distinct child targets, transported by an explicit finite physical protocol, separated after common nuisance completion at a derived finite exposure, and accompanied by a predeclared held-out falsifier.

15 Experimental falsifier and measurement strategy

Definition 15.1 (Frozen falsifier). Let $\widehat{\mathcal{R}}_{\star\eta_d, \eta_q, \alpha_\star}^{\text{strat}}$ be the certified detector-level stratified confidence relation. The falsifier set is

$$\text{Fals}_{p_\star} = \text{QRec}_{p_\star} \setminus \widehat{\mathcal{R}}_{\star\eta_d, \eta_q, \alpha_\star}^{\text{strat}},$$

with source cells, analyzer cells, exposures, calibration, background, reconstruction, confidence coverage, and uncertainty rules frozen before data.

Theorem 15.2 (P7-25 – falsifier theorem). *A complete $p_\star$ record falsifies the PCR-7 prediction lock at the preregistered significance rule when any of the following occurs outside the certified confidence envelope:*

1. *the active-deficit and heavy-weight matrices fail the equality relation;*
2. *the heavy weight is not positive definite of rank three;*
3. *no heavy pole occurs in the scanned interval despite the P7-S1 acceptance and P7-S2A/P7-S2 sensitivity certificate;*
4. *the reconstructed diagonal weights are not flavor universal;*
5. *the record confidence set intersects only the no-vectorlike reference class;*
6. *the stratified source/analyzer response is nonidentifiable, the one-sided quadratic null certificate fails, or an unregistered coherent source is required, in which case the protocol/source certificate is revised rather than the record silently fitted.*

The failure localizes to the observable projection, the source-qualified child, the physical protocol, or an explicit target/source update according to the associated trace coordinate.

Proof. Each listed condition is a defining coordinate of $\mathcal{R}_\star$ or a completeness condition of the physical source-to-record theorem and is retained by the record quotient. Its failure places the complete confidence record outside the predicted class or invalidates the protocol certificate. P7-04/P7-06 localize whether the disagreement arises before or after detector transport. P7-S1, P7-S1A, P7-S1B, P7-S2A, and P7-S2 separate a true packet failure from missing acceptance, missing stratified identifiability, failure of the quadratic null bound, or an unmodelled source. Such an omitted source is an update rather than a silent repair. $\square$

Theorem 15.3 (P7-26 – experiment map). *The preferred implementation class is tagged- W^* flavor-complete charged-current spectral tomography. Its finite order is:*

1. *form a tagged charged-current parent-current source and freeze the source-count normalization;*

2. calibrate charged-lepton flavor tags, recoil/missing-mass response, and control backgrounds;
3. acquire the three one-current light/heavy missing-mass spectra over the complete heavy interval;
4. acquire all ordered-flavor two-current decay/scattering analyzer cells, quadratic-rate controls, and one-sided null-certification sidebands;
5. use sidebands and control samples to determine efficiency, background, and response covariance;
6. reconstruct diagonal and quadratic Gram coordinates, the induced positive-semidefinite matrix/rank class, the pole class, and the simultaneous confidence set in the frozen quotient;
7. compare the complete packet with $\widehat{\mathcal{R}}_{\star, \eta_d, \eta_q, \alpha_\star}^{\text{strat}}$ and $\widehat{\mathcal{R}}_0^{\text{strat}}$ without retuning;
8. optionally add independent LNV and CLFV channels as branch and consistency tests.

No stage may use held-out records to retune I_E , δ_E , the source/analyzer cells, exposure allocation, nuisance model, response class, confidence level, or falsifier.

Proof. The order follows the source-to-record factorization of P7-S1 and the nonlinear finite confidence construction of P7-S2A/P7-S2. Tagged source and calibration records are formed before signal reconstruction. The one-current spectra provide diagonal and pole information; the two-current cells bound the quadratic off-diagonal invariants and close the matrix/rank confidence relation; controls determine the covariance and background; the joint quotient performs the target comparison. LNV/CLFV are not required for the common carrier/reference separation. The data firewall forbids retuning. $\square$

Protocol/channel	Directness	Nondecoupling	Held-out	Authority
Tagged-W active/heavy tomography	High	Physical source and finite margin proved	Future	Cross-child joint-packet lock
One-current heavy pole scan	High	Positive on fixed child, incomplete alone	Future	Conditional without matrix closure
Two-current flavor analyzer	High	Positive sensitivity in all flavor cells	Future	Matrix/rank and universality closure
Active normalization only	High	Fixed δ_E	Partly target-used	Derived relation, calibration-sensitive
Independent LNV search	Medium	Branch-dependent	Future if new protocol	One-sided branch falsifier
CLFV search	Medium	Heavy part exactly zero	Existing/future	Consistency, not separation
Mass ordering response	Medium	Realization route absent	Existing/future	No pre-lock prediction
D1/D4 mass response	Low/medium	Variable/target-used	No	Input, not validation

15.1 Localization of a failed prediction

A failed packet coordinate does not automatically identify which upstream theorem is false. PCR-7 therefore records the following localization map:

Observed discrepancy	Primary theorem under pressure	Alternative update route
$\Delta_{\text{act}} = 0$ with complete sensitivity	Fixed positive s_E / active block	Child source certificate revised.
$\Omega_H = 0$ but $\Delta_{\text{act}} \neq 0$	Vectorlike light-heavy closure identity	Different nonunitary carrier/source.

Observed discrepancy	Primary theorem under pressure	Alternative update route
Heavy pole absent	PCR-5E/PCR-6 heavy sector or P7-S1 coverage	Source/target update or protocol failure.
Pole rank not three	Heavy-degenerate sector theorem or P7-S1B rank certificate	Split/nondegenerate heavy extension or protocol revision.
Flavor weights unequal	Common unitary block factorization	Flavor-nonuniversal source update.
Nonzero heavy CLFV residue	Exact degeneracy/unitary GIM descent	Heavy splitting or new CLFV source.
Packet correct only in one child	Common-core factorization	Decoration backreaction/update.
Unidentified stratified response direction	P7-S1A response design	Additional analyzer/control channel or revised quotient required.

15.2 Predeclared experimental specification

The preregistration contains the target child tags, charged-current source/analyzer normal form, finite mass-window cover, source and control exposures, calibration and sideband plan, record quotient, confidence level, covariance transport, uncertainty budgets, predicted class, reference family, nuisance model, and falsifier set. A later implementation may improve sensitivity but may not change these fields without issuing a new prediction version.

16 Held-out validation and interpretation

Theorem 16.1 (P7-27 – held-out validation theorem). *The PCR-7 prediction is validly pre-registered before empirical comparison. A later data set may be used as validation only if:*

1. *its protocol identifier and response map match $p_\star$ or an explicitly certified refinement;*
2. *it was absent from $\mathcal{D}_{\text{anchor}} \cup \mathcal{D}_{\text{cal}} \cup \mathcal{D}_{\text{derive}}$;*
3. *the uncertainty and nuisance model was frozen before unblinding;*
4. *the comparison reports membership in $\widehat{\mathcal{R}}_{\star, \eta_d, \eta_q, \alpha_\star}^{\text{strat}}$ or $\text{Fals}_{p_\star}$ without retuning.*

At publication, the validation status is

FrozenPredictionAwaitingHeldOutData.

Proof. The data-status ledger fixes all existing records and assigns the primary validation protocol to the future class. The prediction, separator, and falsifier are fully stated without those data. The four conditions are exactly the nonleakage and protocol-identity conditions required for post-lock comparison. Since no qualifying data are imported here, the theorem establishes readiness, not confirmation. $\square$

Warning 16.2 (No retrospective validation). Reanalysis of a D3 channel already used to certify the vectorlike child may test consistency or improve calibration, but it cannot count as independent validation of the same response coordinate unless the new protocol measures a predeclared relation not used in target formation.

16.1 Interpretation of validation outcomes

After a qualifying held-out data set is acquired, the allowed report statuses are:

1. ConsistentWithinEnvelope;
2. FalsifiedObservableProjection;

3. ProtocolIncomplete;
4. SourceOrTargetUpdateIndicated;
5. Indeterminate because the certified uncertainty threshold was not reached.

A sensitivity limit that does not reach the frozen envelope boundary is not a falsification and not a confirmation.

Criterion 16.3 (Blind comparison). Where practical, nuisance and calibration determinations must be frozen using control data that do not reveal the final joint packet classification. Otherwise the data set is calibration/derivation material and a new held-out set is required.

17 Consequences for the PCR-8 capstone

Theorem 17.1 (P7-28 – PCR-8 handoff and finality). *PCR-7 exports the target-tagged object*

$$H_7 = (T_{\text{obs}}^{\nu,b}, p_\star, O_\star^{\min}, \mathcal{R}_\star, \mathcal{R}_0, \text{SrcAna}_\star, \text{Cov}_\star, \text{Sep}_\star, \text{NonDec}_\star, \text{Fals}_{p_\star}, \text{DataStatus}, \partial\text{Fail})_{b \in \mathcal{B}_7^{\text{ready}}}$$

with status `CrossChildJointPacketPredictionLockCCST`, protocol-existence status `FinitePhysicalProtocolRealized`, and validation status `AwaitingHeldOutData`. PCR-8 may use this object to formulate the global capstone but may not extend it to unresolved parent sectors, advertise current facility completion, or treat future validation as already positive.

Proof. The target, packet, and reference are established by P7-00–P7-17. P7-S1 constructs the physical charged-current source/analyzer protocol and its uniform acceptance. P7-S1A establishes stratified diagonal/quadratic/pole identifiability, P7-S1B supplies rank and matrix-space certification, and P7-S2A/P7-S2 construct the finite nonlinear confidence relation and certified margin. P7-19–P7-23 establish separation, nondecoupling, and the prediction lock. P7-25/P7-27 provide the frozen falsifier and validation status. Each component retains its child, protocol, and confidence tag, and the parent firewall remains part of the export. $\square$

Status 17.2 (Finality). The PCR-7 scientific calculation is complete at the declared two-child, source-qualified, finite-protocol scope. The existence and in-principle completeness of the charged-current protocol are proved. Facility realization, exposure accumulation, and held-out empirical comparison remain future operations and are not missing proof premises.

17.1 Scope of the capstone consequence

The capstone may combine the PCR-7 prediction lock with the PCR-6 carrier theorem only on the same two completed child sectors. The resulting target-qualified statement is

$$\text{OneBearer}_{\text{VJ}} + \text{FinitePhysicalProtocol}_{\text{CCST}} + \text{CrossChildPredictionLock}_{\text{CCST}},$$

It does not imply empirical confirmation, current-facility sensitivity, unique microscopic attribution, completion of the unresolved parent neutrino problem, or cross-fermion universality. Those conclusions require, respectively, held-out records, an experimental realization certificate, additional source discrimination, parent closure, or the FCGS programme.

18 Robustness under same-scope counterexamples

A same-scope counterexample must preserve the two completed child targets, the source-qualified E-H2 comparison class, the PCR-6 one-bearer/four-sector theorem, the independently specified reference, the charged-current process class, the finite response model, and the detector-record quotient.

Potential counterexample	Required demonstration and disposition
Data leakage	Exhibit a validation datum used to form the packet, response design, or precision threshold. P7-02 and the provenance ledger exclude this.
Incomplete observable atlas	Exhibit a same-scope semantic use outside the D1–D5/cross-channel grammar. P7-04/P7-05/P7-09 close this at finite source scope.
Reference tuned to result	Show that a reference parameter was chosen after seeing $\mathcal{R}_\star$. P7-10 fixes the source and detector first.
Unphysical protocol	Show that a source/analyzer cell is not generated by the charged-current interaction. P7-S1 gives the one- and two-current amplitudes and their crossed realizations.
Acceptance assumed rather than derived	Exhibit a heavy-threshold point with zero physical response in every declared detector cell. P7-S1 proves positive response at each point and a uniform compact minimum.
Response nonidentifiability	Exhibit a nonzero diagonal, quadratic, or pole coordinate in the null space of all source/analyzer cells. P7-S1A proves finite response injectivity on the stratified target quotient; amplitude tangents at quadratic nulls are intentionally not retained coordinates.
False linearization at a null	Infer an off-diagonal amplitude from a nonzero first derivative of a count rate at zero amplitude. P7-S1A instead uses the invariant coordinate $q = \Omega_{\alpha\beta} ^2$, and P7-S2A derives the corresponding $O(\delta_E^{-4})$ finite-exposure bound.
Uncalibrated systematic margin	Exhibit a retained nuisance without a control sample or covariance coordinate. P7-S1A/P7-S1B/P7-S2A and P7-S1/P7-S2/P7-13 give finite controls, sidebands, stratified confidence transport, and error budgets.
Unitary/basis dependence	Find a lawful U change that alters the packet. Gram-matrix construction makes it invariant.
Small-coupling decoupling	Produce a lawful $s_E \rightarrow 0$ path inside the ready child. P7-12/P7-14 show it leaves the child source certificate.
Heavy-mass decoupling	Send H_λ outside detector reach while preserving protocol. I_E is compact and P7-S1 covers it.
Detector hiding	Let acceptance vanish or add an unmodelled coherent background as nuisance. Such points contradict P7-S1 or update the protocol/source class.
Calibration mimicry	Mimic active deficit by normalization. The minimal packet also requires matched heavy weight, full flavor transfer, and a rank-three pole.
Background pole mimicry	Add a coherent pole to the reference. This is new source content and not a lawful reference presentation.
Branch dependence	Show accidental/exact-global decoration changes the packet. P7-15 proves factorization through the common core.
PCR-5E quotient collapse	Infer one realization from one packet. The theorem expressly denies the converse.
Current-detection overclaim	Treat finite protocol existence as present facility completion or observed data. P7-R5/P7-27 block this.

Theorem 18.1 (Robustness theorem). *Within the declared scope, every same-target attack on the prediction lock reduces to failure of at least one of: upstream child identity, observable completeness, reference completeness, physical source formation, uniform acceptance, response identifiability, covariance/nuisance completion, invariant packet calculation, reference separation, nondecoupling, or data-firewall closure. No independent same-scope selector or hiding route remains.*

Proof. The lock is defined by the conjunction of those predicates. P7-00–P7-12 close target, source, observable, reference, and parameter formation. P7-S1A/P7-S1B/P7-S2A and P7-S1/P7-S2/P7-13 close physical protocol realization, nonlinear null certification, and finite margin. P7-15–P7-21 close invariant content, reference separation,

and nondecoupling. P7-02 closes the data firewall. Fixed-domain exhaustion blocks a further same-scope authority locus. A proposed route that changes source, target, detector principle, reference source, or data role is an update. $\square$

19 Conditions for extension or revision

The result would require revision only if a well-typed comparison altered one of its load-bearing objects. The principal conditions are:

1. a revised PCR-5E/PCR-6 child certificate that changes or removes the fixed positive s_E relation;
2. a lawful ready-child continuation reaching $s_E = 0$ without source/target update;
3. a charged-current source calculation showing that one of the P7-S1 source/analyzer cells is physically absent or has zero response throughout a required flavor/threshold region;
4. a detector-response counterexample showing that the P7-S1A stratified design has a target-relevant diagonal, quadratic, or pole null direction, or that its one-sided quadratic bounds cannot certify the matrix/rank packet after the declared controls;
5. a calibration/background covariance term omitted from the finite nuisance model that can close the carrier/reference gap without source update;
6. a same-reference source producing the rank-three heavy pole and matched nonzero closure packet without adding carrier content;
7. a lawful nuisance direction that closes the carrier/reference gap inside the physical protocol certificate;
8. a provenance audit showing held-out-data leakage;
9. a broader parent-sector theorem with a different carrier core or prediction class.

None is established within the present theorem system. A present facility's inability to deliver the specified exposure does not refute the in-principle finite-protocol theorem; it concerns experimental realization rather than logical validity. Conversely, a null result under a protocol that omits a required source/analyzer cell or fails its calibration/coverage certificate does not falsify the prediction.

Nonclaim 19.1 (No permanence beyond scope). PCR-7 does not assert that every future extension of the neutrino target must preserve the same packet. It asserts the lock for the two completed child targets and the stated source-authorized physical protocol/reference class.

20 Conclusion

PCR-7 establishes an experimental prediction lock for the two neutrino sectors that have reached simultaneous carrier closure and physical-nature classification. The result is not based on a benchmark choice, a projected sensitivity curve, or reuse of an upstream measurement as its own confirmation. It follows from an exact charged-current closure relation that is natural over both completed child groupoids and from a finite source-to-record protocol specified before held-out validation.

The accepted carrier has light and heavy charged-current blocks $N = c_E U$ and $V_H = -s_E U$. Their Gram matrices obey

$$\mathbf{1}_3 - NN^\dagger = V_H V_H^\dagger = \delta_E \mathbf{1}_3, \quad \delta_E > 0.$$

The heavy mass block contributes one threefold-degenerate pole in a compact target-fixed interval, while the heavy CLFV contribution vanishes by row orthogonality. These statements are invariant over the continuum PCR-5E moduli and independent of the accidental or exact-global target decoration.

The detector analysis respects the actual nonlinear geometry of charged-current count records. Light and heavy diagonal weights are inferred from normalization and inclusive-production channels with ordinary two-sided errors. Off-diagonal nulls are represented by the invariant quadratic coordinates

$$q_{\alpha\beta}^L = |(NN^\dagger)_{\alpha\beta}|^2, \quad q_{\alpha\beta}^H = |(V_H V_H^\dagger)_{\alpha\beta}|^2,$$

and certified by one-sided upper bounds, without a singular amplitude-level inversion at zero. The finite source, control, sideband, and mass-response system is injective on this stratified quotient. Predeclared finite exposures give diagonal errors below $\delta_E/8$, quadratic null bounds below $(\delta_E/16)^2$, and retention of the heavy-pole class at the stated simultaneous coverage. The required scaling is $O(\delta_E^{-2})$ for diagonal coordinates and $O(\delta_E^{-4})$ for null-amplitude certification.

The quadratic bounds imply off-diagonal magnitude bounds, and Gershgorin's theorem certifies a positive rank-three heavy matrix. The resulting matrix-packet confidence class lies within radius $3\delta_E/8$ of the ideal carrier class. The independently formed no-vectorlike references contain neither active deficit, heavy spectral weight, nor a heavy pole. Their confidence classes therefore remain separated from the carrier class by

$$\sqrt{6}\delta_E - \frac{3\delta_E}{4} > 0,$$

with rank and pole support providing additional discrete separation. The only algebraic route to the reference would require $s_E \rightarrow 0$, which leaves the source-qualified child certificate; the separation is consequently nondecoupling at the declared scope.

The final scientific status is

CrossChildJointPacketPredictionLock_{CCST}

with protocol status

FinitePhysicalProtocolRealized

and empirical status

FrozenPredictionAwaitingHeldOutData .

The theorem establishes the existence and completeness of a finite physical measurement protocol with a positive pre-data margin. It does not claim current facility realization, completed exposure, or empirical observation.

The exact lower results remain part of the conclusion. The D1/D4 mass records are inputs rather than new predictions; the same-isotope D2 response is not held out; the heavy CLFV null overlaps the reference; and mass ordering remains without a pre-lock prediction. A future nonzero independent LNV result can exclude the exact-global child, whereas a null result alone does not select between the completed sectors. These limitations prevent the charged-current lock from being generalized to unrelated channels.

The PCR-8 capstone can now combine the one-carrier theorem with the finite-protocol prediction lock at the same two target-qualified sectors. Empirical evaluation requires an implementation of the predeclared tagged- W^* protocol, or a certified equivalent, together with the specified exposures, controls, covariance model, and held-out comparison. Any departure from the frozen packet is then localized as a failure of the carrier source, observable transport, protocol certificate, or declared target scope rather than absorbed by retuning the prediction.

A Notation and status calculus

Symbol/status	Meaning
$\mathcal{B}_7^{\text{ready}}$	Two completed vectorlike-joint child scopes.
$X \in \text{PhysMod}_{E,\text{VJ}}^b$	Accepted PCR-5E/PCR-6 realization in child b .
N, V_H	Light and heavy charged-current blocks.
s_E, c_E, δ_E	Fixed child source coefficients with $\delta_E = s_E^2$ and $c_E^2 = 1 - \delta_E$.
Δ_{act}	Active-closure deficit matrix $\mathbf{1} - NN^\dagger$.
Ω_H	Inclusive heavy spectral-weight matrix $V_H V_H^\dagger$.
$p_\star$	Flavor-complete charged-current spectral-tomography protocol.
Ξ_{strat}	Diagonal, squared-modulus, and pole-support detector quotient coordinate.
η_d, η_q	Two-sided diagonal tolerance and one-sided squared-modulus null tolerance.
$O_\star^{\text{min}}$	Minimal packet $(\Delta_{\text{act}}, \Omega_H, \text{Pole}_H)$.
$\mathcal{R}_\star$	Common carrier response class.
$\mathcal{R}_0$	No-vectorlike reference response class.

Symbol/status	Meaning
$\varepsilon_\star$	Certified set-valued product-matrix confidence radius, equal to $3\delta_E/8$ in the finite-exposure construction.
PredLock	Invariant, separated, nondecoupling, protocol-complete, nonleaking prediction status.
AwaitingHeldOutData	Prediction fixed; empirical validation not yet performed.
TargetUpdate	Richer source/target/reference/protocol change requiring a fresh certificate.
NotReady _{κ}	Named missing protocol, response, reference, nuisance, or provenance burden.

B Empirical provenance table

Record family	Role	Reason
Oscillation source/detection evidence	Anchor	Forms PCR-1/PCR-5B target.
KATRIN endpoint release	Anchor/calibration	Forms D1 target and response.
Combined ^{76}Ge LNV response	Anchor/calibration	Forms D2 target.
D3 heavy-state, active closure, LFU, CLFV records	Anchor/derivation	Forms D3-1.3 target and product-box pass.
D4 public mass-sum record	Anchor	Forms public-summary child.
D5 symmetry/charge equations	Derivation/structural	Forms accidental/exact-global decorations.
PCR-5E/PCR-6 source coefficients and intervals	Derivation	Fix s_E , I_E , and common carrier core.
Existing electron active tester	Derivation and calibration	Used upstream; not held out.
Future flavor-complete active/heavy packet	Future held-out	Primary PCR-7 validation.
Independent future LNV protocol	Future held-out	One-sided branch discriminator.
Future complete-cosmology likelihood	Future/NotReady	Outside current D4 child.

C Physical protocol normal form

The finite charged-current spectral-tomography protocol is represented by

$$p = (\text{Parent}, J_Q, \mathcal{K}, \text{Light}, \text{Heavy}, \text{Corr}, \text{FlavorTag}, \text{MassRec}, \text{Cal}, \text{Bkg}, \text{Control}, \text{Quot}, \text{Exposure}, \text{Confidence}, \text{Status}).$$

For the completed protocol $p_\star$ the fields have the following meanings.

Field	Required content
Parent, J_Q	A tagged charged-current parent instrument with kinematic support above the entire heavy interval.
$\mathcal{K}$	A finite cover of compact production, propagation, decay, fiducial, mass-response and control cells.
Light	Flavor-resolved light normalization and seven-point source/analyzer records determining d_α^L and $q_{\alpha\beta}^L = (NN^\dagger)_{\alpha\beta} ^2$ at the declared source-qualified quotient.
Heavy	Inclusive heavy-production records for all three charged-lepton boundary flavors determining $d_\alpha^H = (\Omega_H)_{\alpha\alpha}$ across the complete heavy interval.
Corr	Every applicable flavor- and charge-tagged heavy production-decay or production-scattering record required to measure $q_{\alpha\beta}^H = (\Omega_H)_{\alpha\beta} ^2$.
FlavorTag, MassRec	Flavor identification and calibrated heavy-pole reconstruction with positive response on every applicable cell.
Cal, Bkg, Control	Common carrier/reference flux, efficiency, fiducial, mass-response, background-sideband and reconstruction controls.
Quot	The lawful stratified record quotient preserving diagonal Gram coordinates, quadratic off-diagonal invariants, the induced positive-semidefinite matrix/rank class, and pole support.
Exposure	Finite predeclared signal, calibration, background, response-control and pole-support sample sizes.
Confidence	Two-sided diagonal intervals, one-sided quadratic null bounds, pole support, simultaneous coverage $1 - \alpha_\star$, and induced packet radius $3\delta_E/8 < \sqrt{6}\delta_E/2$.
Status	PhysicalProtocolRealized, ProtocolIncomplete, NotReady, Inapplicable or TargetUpdate.

Criterion C.1 (Protocol completion). The status `PhysicalProtocolRealized` requires all of the following:

1. every signal and control record is the image of a source-coherent physical process;
2. the heavy interval is covered by finitely many positive-response cells with one derived lower acceptance $a_\star > 0$;
3. the carrier and reference share one response, calibration, background and record quotient;
4. the response is injective on the frozen stratified carrier/reference quotient and equipped with finite inverse constants in the diagonal and quadratic invariant coordinates;
5. finite predeclared sample sizes give diagonal $O(\delta_E^{-2})$ and quadratic-null $O(\delta_E^{-4})$ precision, rank/pole certification, and packet radius $3\delta_E/8$ at coverage $1 - \alpha_\star$;
6. source geometry, response, nuisance class, stratified reconstruction relation, one-sided confidence rule and falsifier are frozen before signal-region unblinding.

Failure of any item gives `NotReadyprotocol`, inapplicability or target update. It does not permit positive acceptance, identifiability or precision to be inserted as assumptions after the result is inspected.

Nonclaim C.2 (No present-facility assertion). Protocol completion is an existence theorem at the source-qualified physical comparison scope. It does not assert that an existing experiment has already implemented the full flavor-complete design, accumulated the required finite exposure, or certified the required common covariance.

D Response-envelope normal forms

The ideal carrier and reference envelopes are

$$\mathcal{R}_{\text{car}}^{\text{ideal}} = \{(\delta_E I_3, \delta_E I_3, \text{Deg}_3(H)) : H \in H(I_E)\},$$

$$\mathcal{R}_{\text{ref}}^{\text{ideal}} = \{(0, 0, \text{NoHeavyPole})\}.$$

Let $\mathcal{D}_{p_\star}^{(n_\star)}$ denote the common finite-exposure source-to-record relation of P7-S1A/P7-S1B/P7-S2A and P7-S1/P7-S2. The reconstructed carrier and reference confidence classes are defined by two-sided diagonal intervals, one-sided quadratic off-diagonal bounds, and pole-support predicates. Each continuous matrix packet is contained in a product Frobenius ball of radius

$$\varepsilon_\star = \frac{3\delta_E}{8}$$

about its ideal class. Consequently

$$\text{dist}(\widehat{\mathcal{R}}_{\text{car}}^{1-\alpha_\star}, \widehat{\mathcal{R}}_{\text{ref}}^{1-\alpha_\star}) \geq \sqrt{6}\delta_E - \frac{3\delta_E}{4} > 0.$$

The pole/rank predicates provide an additional discrete separator. These are predicted confidence classes, not empirically observed regions.

Envelope	Free coordinates	Invariant content
Carrier common packet	$H \in H(I_E)$	$\Delta_{\text{act}} = \Omega_H = \delta_E I_3$, positive-definite rank-three class and one degenerate heavy pole.
Accidental decoration	z in the accepted interval	Optional LNV response; no change to the common packet.
Exact-global decoration	fixed $z = 0$	Protector and defined LNV zero; no change to the common packet.
No-vectorlike reference	common protocol nuisances	zero active deficit, zero heavy spectral weight and no heavy pole.
Finite-record carrier class	finite count/control fluctuations	Diagonal intervals, quadratic-null upper bounds, pole support and coverage $1 - \alpha_\star$.
Finite-record reference class	same response and confidence rule	Disjoint from the carrier class by continuous and pole/rank separators.

E Source and dependency index

ID	Manuscript	Role
P7-S01	A Fixed-Domain Exhaustion Theorem	Scope, fail-closed typing, no hidden selector.
P7-S02	Role-Occupancy Closure under Admissibility	Role, identity, and compatibility separation.
P7-S03	Standing Carrier Identity Closure	Stable record/carrier quotient discipline.
P7-S04	Constraint-Induced Physical Carrier Realization ARC Plan	PCR-7 observable and prediction-lock mandate.
P7-S05	Target-Fixed Physical Role Necessity for the Realized Neutrino Oscillation Interface	Empirical anchors and carrier-neutral target.
P7-S06	Complete Physical Interface of the Neutrino Propagation Role Complex	Boundary/evaluator completeness.
P7-S07	Source-First Exhaustion of the Physical Carrier Universe	Source/protocol completeness pattern.
P7-S08	Simultaneous Physical Constraint Collapse of the Neutrino Carrier Universe	One-apex physicality.
P7-S09	Compatible Role-Occupancy Closure for the Neutrino Propagation Role Complex	Cross-realization over-residue control.
P7-S10	Exact Standing-Carrier Stratification of the Neutrino Oscillation Realization Space	Continuum base and matching/threshold standing.

ID	Manuscript	Role
P7-S11	Functorial Target Extension of Standing-Carrier Quotients	Restriction, fibres, no merger, path equivalence.
P7-S12	Direct Kinematic-Mass Target Extension	D1 response and input boundary.
P7-S13	Lepton-Number-Violating Target Extension	D2 amplitude/detector semantics.
P7-S14	Heavy-State, Active-Closure, and CLFV Extension	D3 response and decoupling controls.
P7-S15	Successor-Target Quantitative Active Closure	D3-1.3 tester and target separation.
P7-S16	Cosmological Population and Transport Extension	D4 public-summary input and NotReady boundary.
P7-S17	Symmetry and Charge Extensions	D5 accidental/exact-global decoration.
P7-S18	Joint Standing-Carrier Closure for the Neutrino Sector	Ready children, vectorlike source, moduli and coordinate relations.
P7-S19	Physical Nature and Minimal Identity of the Neutrino Carrier	One bearer, 3 + 1 sectors, observable downstream interface.
P7-S20	Neutrino Ordering Prediction-Lock Closure	Separation/selection and no-lock control.
P7-S21	The Neutrino-Mass Low-Energy Exact-Character Endpoint	Operator occupancy and low-energy scope.
P7-S22	Mass-Matrix, Eigenstructure, and Ratio Rigidity	Basis-invariant spectral discipline.
P7-S23	Mixing-Angle and Phase Standing	Boundary reconstruction and invariant mixing.
P7-S24	The Higgs Domain Endpoint	Downstream projection firewall.
P7-S25	PCR-5E D3 Successor-Branch ARC Target-Update Memorandum	D3-1.3 child authority.
P7-S26	Atre–Han–Pascoli–Zhang, The Search for Heavy Majorana Neutrinos	Charged-current HNL production and decay response.
P7-S27	Abada–Escribano–Marciano–Piazza, Collider Searches for Heavy Neutral Leptons	General HNL collider process and interference structure.
P7-S28	ATLAS prompt HNL search in W decays	Tagged charged-lepton and prompt-decay process realization.
P7-S29	ATLAS displaced-vertex HNL search in W decays	Long-lived propagation, vertex, mass and control-region process realization.
P7-S30	Hoeffding, Probability Inequalities for Sums of Bounded Random Variables	Finite-exposure simultaneous concentration bound.

F Theorem dependency index

ID	Subject	Principal dependencies
P7-00	Import and target lock	Upstream target and preceding local theorems
P7-01	PCR-6 downstream interface finality	Upstream target and preceding local theorems

ID	Subject	Principal dependencies
P7-02	Empirical firewall	Upstream target and preceding local theorems
P7-03	Observable target	Upstream target and preceding local theorems
P7-04	Semantic observable uses	Upstream target and preceding local theorems
P7-05	Protocol grammar	Upstream target and preceding local theorems
P7-06	Record quotient	Upstream target and preceding local theorems
P7-07	Observable map	Upstream target and preceding local theorems
P7-08	Moduli naturality	Upstream target and preceding local theorems
P7-09	Atlas exhaustion	Upstream target and preceding local theorems
P7-10	Reference formation	Upstream target and preceding local theorems
P7-11	Reference completeness	Upstream target and preceding local theorems
P7-12	Parameter statuses	Upstream target and preceding local theorems
P7-13	Physical detector and nuisance completion	P7-S1A, P7-S1B, P7-S2A, P7-S1, P7-S2 and P7-12
P7-14	Decoupling topology	P7-12, P7-S1A–P7-S2A, P7-S1, P7-S2, P7-13
P7-15	Core/decoration factorization	Upstream target and preceding local theorems
P7-16	Invariant packet	Upstream target and preceding local theorems
P7-17	Selection rules	Upstream target and preceding local theorems
P7-18	Ratio/shape/bounded region	Upstream target and preceding local theorems
P7-19	Physical-record separation	P7-16, P7-S1A, P7-S1B, P7-S2A, P7-S1, P7-S2, P7-13
P7-20	Cross-child separation	P7-15, P7-19, P7-S1A–P7-S2A, P7-S1, P7-S2
P7-21	Nondecoupling	P7-12–P7-20, P7-S1A–P7-S2A, P7-S1, P7-S2
P7-22	Prediction status	P7-S1A–P7-S2A, P7-S1, P7-S2, P7-16–P7-21
P7-23	Finite-protocol prediction lock	P7-02, P7-S1A–P7-S2A, P7-S1, P7-S2, P7-13, P7-16, P7-19–P7-21
P7-24	No-lock exhaustion	Upstream target and preceding local theorems
P7-25	Falsifier	Upstream target and preceding local theorems
P7-26	Experiment map	Upstream target and preceding local theorems
P7-27	Held-out validation	Upstream target and preceding local theorems
P7-28	PCR-8 downstream interface	Upstream target and preceding local theorems
P7-S1A	Stratified quadratic-record identifiability	P7-04–P7-09, physical source/analyzer response and compact controls

ID	Subject	Principal dependencies
P7-S1B	Positive-rank and packet-radius certification	P7-S1A and positive-semidefinite/Gershgorin bounds
P7-S2A	Nonlinear finite-exposure theorem	P7-S1A, P7-S1B and finite concentration bounds
P7-S1	Physical-protocol realization and stratified identifiability	P7-S1A, P7-S1B, PCR-5E/PCR-6 source and process certificates
P7-S2	Finite-exposure certified margin and protocol freeze	P7-S2A, P7-S1A, P7-S1B and P7-S1
P7-R1	Higher-arity packet	P7-09, P7-15–P7-20
P7-R2	Minimal packet	P7-R1, P7-16–P7-20
P7-R3	Reference robustness	P7-10–P7-11, P7-19
P7-R4	Cross-child stability	P7-08, P7-15–P7-23
P7-R5	Claim authority	P7-S1A–P7-S2A, P7-S1, P7-S2, P7-02, P7-23, P7-27

G Supplementary formal representations and reproducibility

The supplementary materials provide formal representations of the objects used in the prediction theorem. They record the target fingerprints, empirical provenance partition, observable atlas, charged-current protocol, response covariance, reference classes, parameter and decoupling structure, falsifier, validation status, and capstone interface. Their role is documentary: they support provenance, reproducibility, and internal consistency but do not replace the semantic completeness, physical-response, identifiability, separation, and nondecoupling proofs.

Supplementary object	Scientific content
Target certificate	Ready child scopes, target fingerprints, source-qualified comparison classes, and unresolved parent boundary.
Empirical provenance record	Classification of prior records as target anchors, calibration inputs, derivation inputs, controls, or future held-out validation.
Observable atlas	Target-generated marginal and higher-arity observable classes, lawful overlaps, and typed complements.
Physical protocol certificate	Charged-current source and analyzer cells, compact heavy-window cover, calibration and sideband controls, stratified record coordinates, and applicability conditions.
Response covariance record	Diagonal and quadratic response coefficients, nuisance coordinates, finite-exposure bounds, simultaneous confidence rule, and matrix/pole reconstruction status.
Reference-theory atlas	Unitary and minimal no-vectorlike reference presentations evaluated in the common instrument and record quotient.
Parameter and decoupling record	Lawful source, threshold, scale, detector, and presentation variations together with their exact dispositions.
Prediction and falsifier record	Predicted packet class, confidence separation, nondecoupling status, failure complement, and held-out validation rule.
Experimental channel map	Source, analyzer, control, calibration, and validation roles for each measurement channel.
Capstone interface	Target-qualified PCR-7 result and the boundaries that must remain explicit in PCR-8.

The distributed formal files are checked for schema consistency, theorem references, target tags, and integrity. Successful consistency checks do not constitute evidence for the physical theorem; they verify only that the supplementary representation matches the published argument.

H Claim and nonclaim register

Statement	Status	Scope
One common active/heavy closure packet	Established	Two ready vectorlike-joint children.
Finite source-authorized CCST protocol exists	Established	In-principle tagged charged-current source, finite response cells and controls.
Uniform positive protocol acceptance	Established	Complete compact heavy interval at fixed positive child coefficient.
Target-relative packet identifiability	Established	Frozen carrier/reference source family, not arbitrary nonunitary matrices.
Finite exposure can certify the margin	Established	Any predeclared α_* ; finite sample sizes derived before data.
Positive physical-record gap under p_*	Established	Simultaneous confidence classes at coverage $1 - \alpha_*$.
No lawful decoupling to reference	Established	Ready child and complete physical-protocol closure.
Cross-child prediction lock	Established	Target-, source-, protocol-, exposure- and confidence-qualified.
A present facility has implemented the complete protocol	Not claimed	Actual implementation remains future experimental work.
Packet empirically observed	Not claimed	Held-out data pending.
Vectorlike source uniquely identified by positive observation	Not claimed	Richer source theories may imitate the same packet.
Target-independent lower bound on δ_E	Not claimed	δ_E is fixed by each source-qualified child certificate.
All neutrino realizations satisfy packet	Not claimed	Parent PCR-5E unresolved.
All fermions share packet	Not claimed	FCGS separate.
Mass ordering predicted	Not established	Current prediction-lock certificate remains negative.
Nonzero LNV forced in accidental child	Not established	Only permitted and branch-dependent.
Complete cosmology predicted	NotReady	JLR-D4-1 absent.

I Observable-use grammar

The following table records the primitive protocol operations, their source type, and their retained record consequence.

Generator	Input sort	Output sort	Failure status
Prepare	Boundary/source state	Tagged incident state	Undefined source or boundary
Interact	Prepared state + process	Response amplitude/kernel	Inapplicable topology
Propagate	Response + continuation	Transported state/process	Continuation failure
Scan	Spectral support	Windowed spectral measure	Out of response scope
Tag	Detector interaction	Tagged raw event class	Efficiency/applicability failure

Generator	Input sort	Output sort	Failure status
Reconstruct	Tagged records	Physical response estimator	Incomplete inversion
Normalize	Response + controls	Quotient-stable ratio/matrix	Missing calibration
Correlate	Finite records + incidence	Joint packet	Incompatible source/protocol
Calibrate	Control data + model	Nuisance orbit/certificate	Calibration NotReady
Transport	Record + lawful update	Continued record	Update or failure
Quotient	Complete detector trace	Stable record class	Trace incomplete

Theorem I.1 (Grammar saturation). *Every finite PCR-7 observable protocol in the declared comparison class is a finite typed term over this grammar, modulo source/protocol isomorphism and lawful record equivalence.*

Proof. This restates P7-05 in constructive normal-form form. Every semantic use has a finite causal and record graph. Topologically sort its operations. Each node enters one of the listed source/output types. Shared or higher-arity nodes are generated by Correlate; lawful changes are Transport/Quotient. Induction on the graph reconstructs the term. Conversely each term is semantically typed. $\square$

I.1 Candidate packet generation

The primary packet is generated by two Prepare–Interact–Propagate–Scan–Tag–Reconstruct branches, one active and one heavy, joined by Correlate over the common source and Normalize over shared controls. This derivation is fixed before the packet is tested for separation.

J Decoupling-route analysis

J.1 Source-level routes

1. $s_E \rightarrow 0$: removes the source-active vectorlike incidence and exits the E-H2 child.
2. $\lambda \rightarrow \partial I_E$: remains inside the scan; no weight change.
3. $\lambda \notin I_E$: target/source update.
4. m_0, τ variation: changes mass-sensitive enriched records only.
5. z variation: changes accidental LNV decoration only.
6. exact-global to accidental change: child target update.

J.2 Presentation routes

Gauge, basis, phase, heavy-subspace, and parameter-chart transformations act by conjugation or relabeling and leave the packet class fixed. They cannot change its rank, scalar Gram form, or pole support.

J.3 Detector routes

1. efficiency reduction within the certified positive interval changes raw counts but not the reconstructed packet;
2. efficiency zero invalidates the protocol;
3. finite smearing inside the response certificate preserves the pole class and uncertainty ball;
4. smearing beyond the certificate is `NotReadyresolution`;
5. background variation inside the frozen model cannot produce the correlated source-matched packet;

6. adding a new coherent background pole is a reference/source update.

Theorem J.1 (Decoupling exhaustion). *The preceding source, presentation, and detector routes exhaust the target-preserving closure relevant to $O_{\star}^{\min}$.*

Proof. P7-14 exhausts lawful source and protocol parameters. The source list enumerates every parameter on which the packet could depend. Presentation routes are the lawful groupoid generators. Detector routes are the nuisance generators of P7-13. Any omitted route changes an object outside those signatures and is an update. Therefore the list is complete. $\square$

K Falsifier localization

Failed record	Carrier/source locus	Protocol locus	Permitted interpretation
Δ_{act} central value	Active block coefficient	Active normalization/calibration	Source coefficient revised or protocol incomplete
Ω_H central value	Heavy CC incidence	Heavy acceptance/reconstruction	Heavy source absent/changed or protocol incomplete
$\Delta_{\text{act}} - \Omega_H$	Full light-heavy unitary closure	Common calibration/correlation	Different source architecture or broken joint response
Rank of Ω_H	Heavy multiplicity/degeneracy	Flavor/rank tomography	Heavy sector split or incomplete flavor coverage
Pole support	Heavy mass block	Mass-window scan/resolution	Threshold outside target, no heavy sector, or protocol failure
Flavor universality	Common unitary block	Flavor tagging/cross-calibration	Nonuniversal source or detector bias
Heavy CLFV zero	Exact degeneracy/GIM	CLFV response model	New CLFV source or heavy splitting
Cross-child equality	Common core	Branch protocol identity	Decoration backreaction or mismatched protocols

Remark K.1. A falsified packet does not automatically falsify the AASC kernel or the entire neutrino programme. It identifies the first load-bearing source, response, or protocol locus at which the frozen chain fails.

L Experimental protocol specification

L.1 Measurement objective

The protocol reconstructs, in one common record system, the active closure deficit, inclusive heavy charged-current weight and heavy-pole support for all three charged-lepton boundary flavors over the complete compact heavy interval. It compares carrier and no-vectorlike reference predictions through the same source, calibration, background, response and record quotient.

L.2 Source and response

The source is a tagged charged-current parent instrument with invariant mass in a compact interval J_Q lying above the heavy-production thresholds. For each $\alpha \in \{e, \mu, \tau\}$ the protocol includes light normalization records and inclusive heavy-production records. Every kinematically applicable correlated heavy production-decay or

production-scattering cell

$$W_\star \rightarrow \ell_\alpha N_\alpha, \quad N_\alpha \rightarrow \ell_\beta X_{CC} \quad \text{or} \quad N_\alpha + T_\beta \rightarrow \ell_\beta + X_\beta$$

is retained. Exact heavy degeneracy makes the coherent amplitude proportional to $(V_H V_H^\dagger)_{\beta\alpha}$, while the tagged probability is proportional to the invariant quadratic coordinate $|(V_H V_H^\dagger)_{\beta\alpha}|^2$.

Subprotocol	Required output	Certificate
Parent controls	Tagged source flux and kinematics	Source authorization, finite parent-energy interval and common carrier/reference instrument.
Light tomography	d_α^L and $q_{\alpha\beta}^L = (NN^\dagger)_{\alpha\beta} ^2$	Flavor tags, normalization controls, seven predeclared phase samples and invertible trigonometric design.
Heavy production	$d_\alpha^H = (V_H V_H^\dagger)_{\alpha\alpha}$ and heavy mass support	Finite cover of $H(I_E)$ and derived acceptance lower bound.
Correlated heavy analyzers	$q_{\alpha\beta}^H = (V_H V_H^\dagger)_{\alpha\beta} ^2$	Flavor/charge tags, common propagator, decay/scattering alternatives and typed N/A for closed cells.
Quadratic null certification	One-sided upper bounds on q^L, q^H	Positive rate coefficients, sideband subtraction and no amplitude-level tangent inversion.
Flavor closure	Positive-definite rank-three class	Diagonal precision and quadratic off-diagonal bounds sufficient for a positive spectral lower bound.
Mass reconstruction	One degenerate pole class in $H(I_E)$	Calibrated finite mass-response cells and background sidebands.
Common calibration	Comparable carrier/reference normalization	Shared controls, covariance and the stratified response relation.
Finite record quotient	$O_\star^{\min}$ confidence class	Frozen set-valued reconstruction, nuisance class, exposure and simultaneous coverage rule.
Blind validation	Membership or falsifier result	Signal-region holdout and no retuning.

L.3 Count and control model

For finitely many signal and control cells, the recorded counts may be written

$$Y_k \sim \text{Poisson}(L_k[\eta_k \mu_k(o_{\text{strat}}) + \beta_k]),$$

with separately tagged calibration and sideband controls for L_k, η_k, β_k and the mass-response parameters. The stratified coordinate vector is

$$o_{\text{strat}} = (d^L, q^L, d^H, q^H, \pi_H).$$

After the controls are ordered first, the quadratic coordinates q enter through linear positive-response blocks, while the diagonal coordinates d are recovered through finite blockwise inverse constants on the compact carrier/reference domain. No globally Lipschitz inverse from $q = |\Omega_{\alpha\beta}|^2$ to a complex off-diagonal amplitude is asserted at $q = 0$.

L.4 Finite-exposure margin

Let M_{diag} and M_{quad} be the finite diagonal and quadratic record dimensions, and let $L_{\text{diag}}, L_{\text{quad}}$ be the corresponding inverse constants. Conservative sufficient bounded-frequency counts are

$$n_{\text{diag}} \geq \frac{128L_{\text{diag}}^2}{\delta_E^2} \log\left(\frac{8M_{\text{diag}}}{\alpha_\star}\right),$$

$$n_{\text{quad}} \geq \frac{131072L_{\text{quad}}^2}{\delta_E^4} \log\left(\frac{8M_{\text{quad}}}{\alpha_\star}\right).$$

Finite calibration, background, response-control and pole-support exposures are added with the same preregistered error allocation. At the resulting simultaneous coverage,

$$|\widehat{d}_\alpha - d_\alpha| < \frac{\delta_E}{8}, \quad q_{\alpha\beta}^U < \left(\frac{\delta_E}{16}\right)^2.$$

The induced positive-semidefinite matrix packet has radius

$$\varepsilon_{\text{strat}} = \frac{3\delta_E}{8} < \frac{\sqrt{6}}{2}\delta_E,$$

and the carrier/reference confidence classes remain disjoint.

L.5 Requirements for experimental realization

The theorem establishes the required physical source, finite response cover, calibrated nuisance model, and finite sample sizes at the accepted source-qualified comparison scope. It does not claim that a current facility already supplies the complete flavor coverage, detector geometry, control statistics, or uncertainty budget. An experimental realization must specify the actual response model, calibration covariance, background class, stratified reconstruction relation, exposure allocation, confidence rule and frozen falsifier.

Warning L.1 (No projected sensitivity as proof). A projected sensitivity curve, informal multiplication of marginal experiments, amplitude-level linearization at a null, or post-data tuning does not discharge the protocol theorem. Loss of physical source coverage, positive acceptance, stratified identifiability, one-sided quadratic null certification, common calibration or the certified confidence margin yields $\text{NotReady}_{\text{protocol}}$ or target update.

M Empirical provenance matrix

Record	First theorem use	PCR-7 use	Held-out?	Backflow prohibition
Oscillation transfer/frequencies	PCR-1	Target identity	No	Cannot validate carrier prediction
KATRIN endpoint	D1	Input/correlation only	No	No same-response prediction claim
^{76}Ge null response	D2	Branch input	No	No same-isotope validation
D3 direct searches	D3	Child formation	No	Cannot validate same H charts
Electron active tester	D3-1.3	Derives δ_E child status	No	Electron value not held out
D3 LFU/CLFV	D3	Child pass/consistency	No	New protocol relation required
D4 public mass sum	D4	Child formation	No	Not a prediction
D5 symmetry/protector	D5	Branch definition	No	Structural use only
Future CCST packet	PCR-7	Validation only	Yes/future	Cannot tune target/protocol
Future independent LNV	PCR-7	Branch test	Yes/future	Cannot select source before lock

Criterion M.1 (Validation integrity). A record changes from future/held-out to calibration or derivation status if any part of it is used to choose the mass window, fit the packet, narrow the nuisance set, set the uncertainty threshold, or alter the falsifier. A new held-out data set is then required.

N Block-unitary derivation of the closure packet

N.1 Exact source rotation

The vectorlike-joint source admits a block-unitary diagonalization of the light/heavy Dirac mass map. At the charged-current boundary, the relevant row is represented by the isometry

$$\mathcal{V}_{\text{CC}} = \begin{pmatrix} c_E U & -s_E U \end{pmatrix} : \mathbb{C}_{\text{light}}^3 \oplus \mathbb{C}_{\text{heavy}}^3 \longrightarrow \mathbb{C}_{\text{boundary}}^3.$$

Its completeness relation is

$$\mathcal{V}_{\text{CC}} \mathcal{V}_{\text{CC}}^\dagger = c_E^2 U U^\dagger + s_E^2 U U^\dagger = \mathbf{1}_3.$$

Writing $N = c_E U$ and $V_H = -s_E U$ yields

$$N N^\dagger + V_H V_H^\dagger = \mathbf{1}_3.$$

The closure packet is therefore a boundary consequence of the exact block isometry rather than an independent phenomenological ansatz.

N.2 Spectral decomposition

The heavy mass block is

$$M_H = H_\lambda \mathbf{1}_3.$$

Let E_H be the heavy eigenspace and P_H its aggregate projector. The spectral measure carried by the charged-current boundary is

$$\mu_H^\alpha = \sum_{k=1}^3 |(V_H)_{\alpha k}|^2 \delta_{H_\lambda} = \delta_E \delta_{H_\lambda}$$

for every boundary flavor α . The total matrix-valued measure is

$$\mu_H = V_H V_H^\dagger \delta_{H_\lambda} = \delta_E \mathbf{1}_3 \delta_{H_\lambda}.$$

The lawful $U(3)_H$ action changes the heavy basis but leaves this measure fixed. The protocol therefore reconstructs one rank-three pole class, not three labeled heavy particles.

N.3 Light closure deficit

The light matrix-valued boundary measure is

$$\mu_L = N N^\dagger = c_E^2 \mathbf{1}_3.$$

Relative to the unitary reference, the missing light weight is

$$\mathbf{1}_3 - \mu_L = \delta_E \mathbf{1}_3,$$

exactly the heavy weight. This is the source-level origin of the detector packet.

Theorem N.1 (Basis-free closure identity). *The equality*

$$\mathbf{1}_3 - N N^\dagger = V_H V_H^\dagger$$

is invariant under all lawful boundary and heavy-subspace unitary transformations and under every parameter variation in the ready child groupoids.

Proof. Boundary transformations conjugate both sides by the same unitary; heavy-subspace transformations cancel inside $V_H V_H^\dagger$; source and enriched parameters other than s_E do not enter; and s_E is fixed by the child certificate. Hence the identity is intrinsic. $\square$

N.4 CLFV residue

For any loop function F common to the degenerate heavy poles,

$$\mathcal{A}_{\beta\alpha}^H \propto F(H_\lambda^2/m_W^2) (V_H V_H^\dagger)_{\beta\alpha}.$$

Thus

$$\mathcal{A}_{\beta\alpha}^H = 0 \quad (\beta \neq \alpha), \quad \mathcal{A}_{\alpha\alpha}^H \propto \delta_E F.$$

The off-diagonal zero is a consistency relation; the diagonal inclusive strength is part of the heavy weight. This clarifies why the CLFV null alone does not separate the reference while the complete packet does.

O Prediction-status matrix

Observable	Invariant?	Reference separated?	Nondecoupling?	Final status
$O_{\star}^{\min}$	Yes, both children	Yes, physical finite-record confidence classes	Yes	Cross-child finite-protocol joint-packet lock
Δ_{act} alone	Yes	Nominally; calibration-sensitive	Fixed source	Derived; insufficient alone
Ω_H alone	Yes	Conditional on source attribution	Fixed source	Derived; insufficient alone
Heavy pole only	Pole window fixed	Yes under P7-S1–P7-S2 with P7-S1A/P7- S1B/P7-S2A	Yes in I_E	Derived; insufficient without joint closure
Heavy CLFV zero	Yes	No	Yes	Structural only; overlapping
Absolute beta scale	No; varies	Target-used	N/A	Input; not held out
Cosmological mass sum	Bounded; varies	Target-used	N/A	Input; not held out
Accidental LNV	Variable	Branch-dependent	May vanish	Conditional observable
Exact-global LNV zero	Exact	Reference overlaps	Yes	Branch-discrimination only
Mass ordering	Candidate responses exist	No realized response	Not closed	No pre-lock prediction

Proposition O.1 (Uniqueness of the positive lock in the current atlas). *At the declared support and protocol scope, $O_{\star}^{\min}$ is the only observable object satisfying all five prediction-lock predicates across both ready children.*

Proof. The table records the complete dispositions proved in P7-22/P7-24. Every other candidate fails invariance, separation, nondecoupling, protocol completeness, or the data firewall. Atlas exhaustion excludes an unlisted same-scope route. $\square$

P Synopsis of principal results

This appendix summarizes the logical order and principal scientific output of the theorem chain.

ID	Result	Scientific content
P7-00	Import/target lock	Exact two-child authority; no parent or downstream backflow.
P7-01	PCR-6 downstream interface finality	Complete carrier, sector, source, and continuation input.
P7-02	Empirical firewall	Used and held-out data are disjoint.
P7-03	Observable target	Branch-indexed target formed without preferred experiment.
P7-04	Semantic observable use	Complete target-facing observable trace.
P7-05	Protocol grammar completeness	Every finite semantic use has a source/protocol normal form.
P7-06	Record quotient	Unique stable detector-record identity.
P7-07	Observable map	Total fail-closed source-to-record projection.
P7-08	Moduli naturality	Lawful groupoid arrows preserve record classes/envelopes.
P7-09	Atlas exhaustion	Complete marginal and higher-arity protocol classes.
P7-10	Reference formation	Independent unitary/minimal no-vectorlike comparators.
P7-11	Reference completeness	Zero active deficit, zero heavy weight, no heavy pole.
P7-12	Parameter status	s_E fixed; other lawful variables classified.

ID	Result	Scientific content
P7-S1A	Stratified quadratic-record identifiability	Linear inversion in diagonal and squared-magnitude invariants; one-sided null certification.
P7-S1B	Rank certification	Diagonal lower bounds plus quadratic null bounds imply rank three and a finite packet radius.
P7-S2A	Nonlinear finite exposure	Correct $O(\delta_E^{-2})$ diagonal and $O(\delta_E^{-4})$ null-amplitude scaling.
P7-S1	Physical protocol realization	Finite source-to-record construction, uniform acceptance, common response, and stratified identifiability.
P7-S2	Finite-exposure certified margin	Finite preregistered signal/control samples, one-sided null bounds, rank closure, and positive detector margin.
P7-13	Detector/nuisance completion	Complete physical protocol and finite calibrated nuisance object.
P7-14	Decoupling topology	All lawful source/protocol hiding routes identified.
P7-15	Core/decoration factorization	Packet common to accidental and exact-global children.
P7-16	Invariant packet	$\Delta_{\text{act}} = \Omega_H = \delta_E I_3$, rank three, degenerate pole.
P7-17	Exact selection rules	Rank, closure equality, and heavy GIM residue.
P7-18	Shape/bounded region	Flavor-universal ratios and closed packet envelope.
P7-19	Branch separation	Exact rank/pole separation and positive matrix gap.
P7-20	Cross-child separation	One protocol and class work for both children.
P7-21	Nondecoupling	No lawful child/protocol path reaches reference.
P7-22	Strength classification	Exact authority for every atlas route.
P7-23	Prediction lock	Cross-child joint-packet prediction theorem.
P7-24	No-lock exhaustion	Exact lower statuses for all other routes.
P7-25	Falsifier	Frozen record complement and failure localization.
P7-26	Experiment map	Staged flavor-complete tomography protocol.
P7-27	Held-out validation	Prediction frozen; future data not yet used.
P7-28	PCR-8 downstream interface	Complete prediction-lock object and open boundaries.
P7-R1	Higher-arity packet	Joint packet irreducible at complete protocol.
P7-R2	Minimal packet	Δ_{act} , Ω_H , and pole support are necessary/sufficient.
P7-R3	Reference robustness	Lawful reference presentations remain in zero class.
P7-R4	Cross-child stability	Common packet/falsifier natural across children.
P7-R5	Claim authority	Exact permissible prediction language.

Proof spine

PCR-6 carrier/source packet

- ⇒ observable atlas (P7-04–P7-09)
- ⇒ reference and physical-protocol closure (P7-10–P7-14; P7-S1, P7-S1A, P7-S1B, P7-S2A, P7-S2)
- ⇒ finite-record separation (P7-15–P7-21)
- ⇒ CrossChildJointPacketLock_{CCST} (P7-22–P7-23)
- ⇒ falsifier and PCR-8 handoff (P7-25–P7-28).

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